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Journal of Anthropology of Sport and Physical Education

ORIGINAL SCIENTIFIC PAPER

Trends in Body Height, Body Weight, and Body Mass Index from 1979 to 1987: An Analysis of the Young Male Population from the Municipality of Cetinje

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Abstract

This study aimed to consolidate body height, body weight, and body mass index data of the entire young male population from the Municipality of Cetinje in order to estimate trends from 1979 to 1987. The sample of respondents includes 2119 young males divided into nine groups: 183 respondents born in 1961, 361 respondents born in 1962, 300 respondents born in 1963, 162 respondents born in 1964, 155 respondents born in 1965, 258 respondents born in 1966, 218 respondents born in 1967, 234 respondents born in 1968, and 248 respondents born in 1969. The measurement sample includes body height, body weight, and body mass index calculated based on two previous measures. The descriptive statistics were expressed as a mean and standard deviation for each variable, while the analysis of nutrition status was done based on body mass index (underweight, normal weight, pre-obese, obese). The results showed that a secular trend in the observed study period is not visible regarding body height and body weight, while it is observable for the body mass index. Therefore, this study will contribute to complementing knowledge about changes in average body height values of young Montenegrins in the previous 120 years and in that way, more precisely monitor the emergence of a secular trend.

Keywords: *Morphological Characteristics, Secular trend, Young males, Montenegro*

Introduction

In recent decades, average adult body height has dramatically increased in most industrialized countries, including in Montenegro (Popovic, 2017). Much better lifestyles, a result of better living conditions and improved nutritional, hygienic, economic, and health status, obviously caused this trend (Hauspie, Vercauteren, & Susanne, 1996; Masanovic, Bavcevic, & Prskalo, 2019a). Trends in men's body height have been analysed around the world for 250 years (NCD Risk Factor Collaboration, 2016), and the unusual height of Montenegrins was recognized by Robert Ehrich, who conducted research at the beginning of the 20th century (Coon, 1939, 1975). Generally, the unusual height of Dinaric Alps inhabitants has historically been well-known (Coon, 1939; Grasgruber et al., 2017, 2019), but the problem is the lack of record-keeping. Recently, studies about body height of Montenegrins have been growing in number; all of them confirm that Montenegrins are one of the tallest nations in the world (Bjelica et al., 2012; Milasinovic, Popovic, Matic, Gardasevic, & Bjelica, 2016; Milasinovic, Popovic, Jaksic, Vasiljevic, & Bjelica, 2016). Nevertheless, between the research and study conducted by Ehrich at the beginning of the 20th century and these recent studies, there is a multi-decade gap of quality data, which will be filled by this study.

Bodyweight and the body mass index (BMI) are parame-

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ters that provide information about nutrition, and it is generally known that being underweight, overweight, or obese is associated with adverse health consequences throughout the life-course (NCD Risk Factor Collaboration, 2017). Thus, both excesses and deficiency of adipose tissue have harmful metabolic consequences and represent significant medical and socioeconomic burdens in the world today (Masanovic, Bavcevic, & Prskalo, 2019b). Underweight among children is associated with a higher risk of infectious diseases, while it may impair reproductive capacity in young people (Han, Mulla, Beyene, Liao, & McDonald, 2010, Masanovic, Milosevic, & Corluka, 2018; Masanovic, Corluka, & Milosevic, 2018). In contrast, being overweight is associated with a greater risk of cardiovascular diseases and chronic disorders, such as type 2 diabetes (Singh, Mulder, Twisk, Van Mechelen, & Chinapaw, 2008).

Although body height, body weight, and body mass index trends in adolescents are documented in the majority of countries, there is insufficient data about trends in Montenegro. There are longitudinal studies about body height, but very few monitor average body weight and body mass index (BMI) changes for a longer period, also those studies that monitor underweight, overweight, or obesity problems. This study brings together the body height, body weight, and body mass index data of an entire young male population of the Municipality of Cetinje, to evaluate the trends from 1979 to 1987, for the purpose of collecting information on possible acceleration, as well the trajectories (changes) of nutrition status in young males.

Methods

The population of this study contains all young male citizens of the Municipality of Cetinje, measured during mandatory medical examinations to test their military service ability. Most future recruits underwent this examination before they were 18 years old, but military service could be postponed until the age of 27, so some of the future recruits had their medical examinations later, which increased the average age in each generation.

In the period from 20 February 1979 to 21 April 1987, this examination consisted of 2146 of future recruits, but young males born in 1957 (3 respondents), 1958 (4 respondents), 1959 (6 respondents) and 1960 (14 respondents) were excluded from the analysis because their numbers were not sufficient for their entire generation's characteristics to be reliably described. Consequently, the analysed data in this study covers 2119 future recruits divided into nine groups: 183 respondents born in 1961 (17.97±.99 yrs.), 361 respondents born in 1962 (17.72±.45 yrs.), 300 respondents born in 1963 (17.89±.48 yrs.), 162 respondents born in 1964 (18.46±.67 yrs.), 155 respondents born in 1965 (18.4±.66 yrs.), 258 respondents born in 1966 (18.4±.37 yrs.), 218 respondents born in 1967 (18.35±.22 yrs.), 234 respondents born in 1968 (17.71±.05 yrs.), and 248 respondents born in 1969 (17.76±.04 yrs.).

Anthropometric measurement was implemented in medical infirmaries, and respondents accessed the procedure in their underwear. From the sample measures that were collected, for this research, body height and body weight are isolated; the body mass index is calculated using them. For body height and body weight assessment, a medical scale with moving weights with a stadiometer was used. Anthropometrical measurement was implemented by respecting the basic rules and principles of the International Biological Program (IPB), and the body mass index was calculated based on the protocol handbook for physical form assessment connected to health (Kaminsky, 2013).

The data obtained in the research were processed using SPSS 20.0 software (Chicago, IL, USA) adjusted for use on personal computers. The descriptive statistics were expressed as a mean and standard deviation for each variable, while the analysis of nutrition status was done based on body mass index (underweight, normal weight, pre-obese, obese) (World Health Organization, 2010).

Results

Analysis of the average body height, body mass, and body mass index of young male subjects is shown in Table 1. The average body height of the overall sample of male subjects was 178.38±6.58 centimetres. The tallest group were respondents born in 1962 (179.76±6.62), while shorter ones were respondents born in 1964 (176.55±6.28). The average bodyweight of the overall sample of male subjects was 70.16±9.17 kilograms, while the heaviest respondents were those born in 1969 (71.74±10.35), and the least heavy were respondents from group born in 1961 (68.91±8.83). The average body mass index of the overall sample of male subjects was 22.02, while the highest values had respondents of the group born in 1969 (22.35), and the lowest values had respondents of the group born in 1965 (21.69).

Table 1. Descriptive data of young male from Cetinje enrolled in the study

Year of birth	1961 (n=183)	1962 (n=361)	1963 (n=301)	1964 (n=162)	1965 (n=155)
	Mean±SD	Mean±SD	Mean±SD	Mean±SD	Mean±SD
Age (yrs.)	17.97±0.99	17.72±0.45	17.89±0.48	18.46±0.67	18.4±0.66
Body Height (cm)	177.93±6.96	179.76±6.62	178.08±6.1	176.55±6.28	179.01±6.99
Body Weight (kg)	68.91±8.83	70.6±8.95	69.25±8.38	69.44±9.71	69.6±8.89
BMI (kg/m^2)	21.74±2.21	21.82±2.27	21.83±2.35	22.26±2.7	21.69±2.23
Year of birth	1966 (n=258)	1967 (n=218)	1968 (n=234)	1969 (n=248)	1961-1969 (n=2119)
	Mean±SD	Mean±SD	Mean±SD	Mean±SD	Mean±SD
Age (yrs.)	18.4±0.37	18.35±0.24	17.71±0.05	17.76±0.04	18.02±.57
Body Height (cm)	179.09±5.98	177.81±6.46	177.01±7.05	178.92±6.45	178.38±6.58
Body Weight (kg)	70.95±8.99	70.49±9.07	69.59±9.3	71.74±10.35	70.16±9.17
BMI (kg/m^2)	22.11±2.49	22.26±2.33	22.17±2.36	22.35±2.56	22.02±2.4

Trends in mean body height, body weight, and body mass index (BIM) by year of birth are presented graphically (Figures 1, 2, and 3).

From Table 2, it can be observed that, in the overall sample of subjects, 4.39% were underweight, 86.27% has normal weight, 8.38% were pre-obese, and 0.9% were obese. The highest percentage of underweight is in the group of respondents born in 1963 (7.61%), while the lowest percentage is in the group of

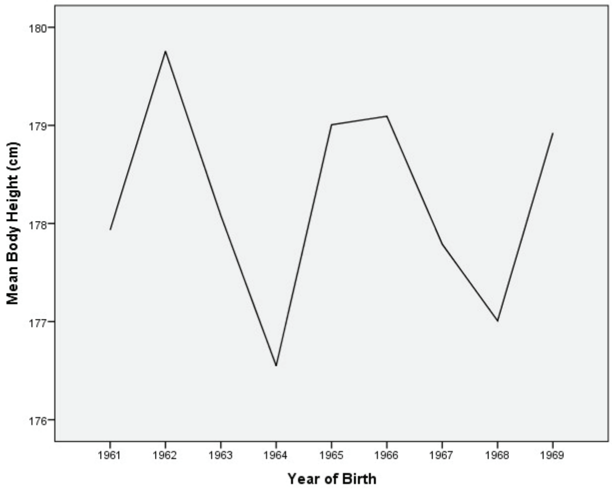

FIGURE 1. Trends in mean body height by year of birth in young males

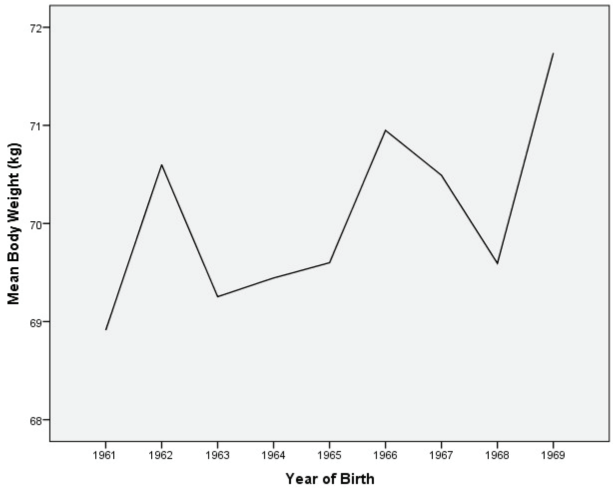

FIGURE 2. Trends in mean body weight by year of birth in young males

FIGURE 3. Trends in mean body mass index (BMI) by year of birth in young males

respondents born in 1967 (2.28%). The highest percentage of respondents with normal body weight is in the group of respondents born in 1962 (91.41%), while the lowest percentage is in the group of respondents born in 1964 (83.33%). The highest percentage of pre-obesity is in the group of respondents born in 1968 (11.97%), while the lowest percentage is in the group of respondents born in 1962 (3.6%). Lastly, the highest percentage of obesity is in the group of respondents born in 1964 (1.85%), while among respondents born in 1961 and 1968, there was not a single obese respondent.

Table 2. The Nutrition status by age and total for the young male enrolled in the study

Year of birth	1961		1962		1963		1964		1965	
	(n=183)	%	(n=361)	%	(n=301)	%	(n=162)	%	(n=155)	%
Underweight	11	6.01	13	3.6	22	7.31	6	3.7	7	4.52
Normal weight	161	87.98	330	91.41	252	83.72	135	83.33	139	89.68
Pre-obese	11	6.01	13	3.6	25	8.31	18	11.11	8	5.16
Obese	0	0	5	1.39	2	0.66	3	1.85	1	0.65
Year of birth	1966		1967		1968		1969		1961-1969	
	(n=258)	%	(n=219)	%	(n=234)	%	(n=248)	%	(n=2119)	%
Underweight	8	3.1	5	2.28	9	3.85	12	4.84	93	4.39
Normal weight	222	86.05	183	83.56	197	84.19	209	84.27	1828	86.27
Pre-obese	25	9.69	26	11.87	28	11.97	23	9.27	177	8.38
Obese	3	1.16	1	0.46	0	0	4	1.62	19	0.9

Trends in all categories of nutrition status (underweight, normal weight, pre-obese, obese) by year of birth are presented graphically (Figure 4).

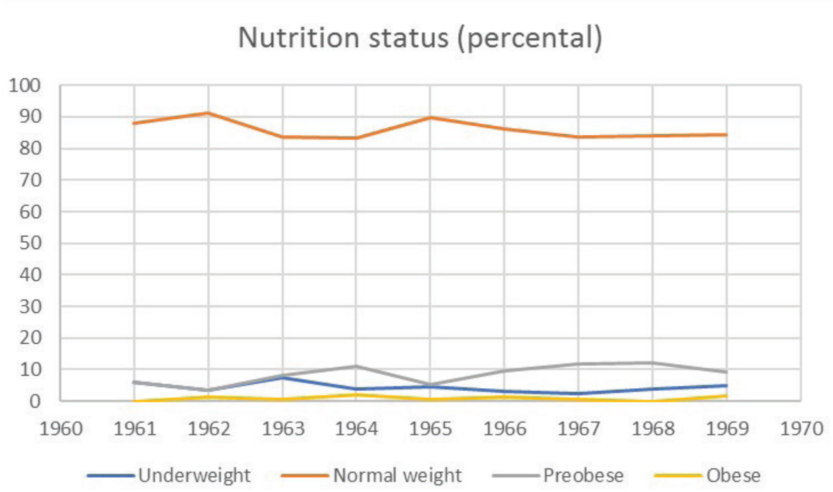

FIGURE 4. Trends in the nutrition status by year of birth in young males

Discussion

This study contributes to the previous knowledge about changes in average body height values of young Montenegrins in the previous 120 years and, in that way, more precisely tracks the secular trend. As in the beginning of the 20th century, the average body height of young Montenegrins, according to Robert Ehrich, was 177 cm (Coon, 1939) and, in the period from 1979 to 1987, young Montenegrins enrolled in this study were 178.38 cm tall, it is possible to conclude that for this period a secular trend is visible, and that body height has increased by 1.38 cm in average. Further, adding this to the results of Popovic (2017), who established that average body height of young male from 13 municipalities all over Montenegro is 183.36 cm, it is clear that the trend in this period is also ascending, and that the height of young Montenegrins increased by 4.98 cm in the last 30 years. For the period from 1979 to 1987 covered by this study, changes are not visible, which is logical, because an eight-year period is too short for a noticeable secular trend.

In accordance with body height changes, other parameters (average body weight, body mass index value, and global obesity frequency) increased for children and adolescents in the previous 40 years (NCD Risk Factor Collaboration, 2017). According to the results of the previously mentioned study, in that period, in 189 countries, the average body mass index value increased for young male respondents by more than 0.05 kg/m² for every 10 years. Furthermore, obesity in last 40 years, for children and adolescents, increased from 0.9% to 7.8%. Lastly, the percentage of underweight for young males decreased from 14.8% to 12.4%.

The present study's results indicate that in the period from 1979 to 1987, the increase in body mass index values of young male Montenegrins is visible, which can be seen in Figure 3. Therefore, respondents born from 1961 to 1965 have lower body mass indexes than respondents born from 1966 to 1969, whose body mass index value is always above 22 kg/m². This result is in accordance with the aforementioned study (NCD Risk Factor Collaboration, 2017), which indicates that obesity frequency is increasing in all countries, even if that increase is no significant in countries with high income. Furthermore, it is interesting to compare the average body mass index of respondents from this study (measured from 1979 to 1987), whose value is 22.02±2.4 kg/m², with respondents measured by Gardasevic et al. (2015), whose average body mass index value was 24.9 kg/m² (17 yrs.) and 22.8 kg/m² (18 yrs.), because can conclude that the secular trend is also evident here.

One limitation of this study is the fact that respondents' average age was 18,02 years, so it can be reasonably concluded, that at the time of measurement, their growth still had not finished. Considering this fact, it is assumed that, their body height data would progress, and would have been higher if they were measured one year later. Of course, then the difference would be greater between them and respondents who were measured at the beginning of the century, which is supported by the fact that the vast majority of young men have completed this examination before the age of 18, which practically means that the vast majority of them were underage at the time of measurement, so it can be said with certainty that they had not reached its final growth. However, military service may have been postponed until the age of 27 of each individual, which some future recruits did, and a medical examination was performed much later, and therefore raised the average age of the entire sample. Consequently, data about their final characteristics are not completely reliable. Certainly, this does not diminish the contribution of this study, which contributes to alleviating the lack of data about trends body height, body weight, and body mass index, bridging the gap of almost 100 years, in which data about such morphological characteristics of young male Montenegrins were not recorded.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Elite Football Players of Bosnia and Herzegovinian and Kosovian Clubs and Differences in the Morphological Characteristics and Body Composition among them

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Abstract

The goal of this research was to determine the differences in the morphological characteristics and body composition among the top football players of the two clubs, Bosnia and Herzegovinian football club Zrinjski Mostar and Kosovian football club Besa Peje. A sample of 46 subjects was divided into two sub-samples. The first sub-sample of the subjects consisted of 28 players of CSC Zrinjski Mostar of the average age 24.36±4.14, the champion of the Bosnia and Herzegovina in the season 2016/17, while the other sub-sample consisted of 18 players of FC Besa Peje of the average age of 21.83±3.17, the winner of Kosovo Cup in the season 2016/17. Football players were tested immediately after the end of the competition season 2016/17. Morphological characteristics in the body composition were evaluated by a battery of 11 variables: body height, body weight, waist circumference, triceps skinfold, biceps skinfold, back skinfold, abdominal skinfold, body mass index, fat percentage and muscle mass. The differences between the players of the top two football clubs in the morphological characteristics and variables for assessing body composition was determined by a t-test for independent samples. It was found that the football players of CSC Zrinjski and FC Besa Peje have statistically significant differences by the three variables that estimate the body weight, waist circumference and muscle mass.

Keywords: *Football Players, Morphological Characteristics, Body Composition*

Introduction

A football game is said to be the most important secondary thing in the world, it gathers huge masses at stadiums and in front of TVs (Bjelica, Popovic, Gardasevic, & Krivokapic, 2016; Sermaxhaj, Popovic, Bjelica, Gardasevic, & Arifi, 2017). It is a highly dynamic and fast team game which, with its richness of movement, falls under category of polystructural sports games (Gardasevic, Georgiev, & Bjelica, 2012; Gardasevic, Bjelica, Milasinovic, & Vasiljevic, 2016). Football is a sport that is characterized by numerous and various complex and dynamic kinesiological activities which are then characterized by either cyclical (Gardasevic, Popovic, & Bjelica, 2016; Gusic, Popovic, Molnar, Masanovic, & Radakovic, 2017) or acyclical movement (Gardasevic, & Vasiljevic, 2017). In football, top score can be achieved only under conditions of well-programmed training process (Gardasevic, Bjelica, & Corluka, 2018). High quality management of the training process depends on the knowing of the structure of certain anthropological capabilities and player's characteristics, as well as their development (Masanovic, 2018; Gardasevic, & Bjelica, 2018; Arifi, Bjelica, & Masanovic, 2019). Various researches are to be done in order to establish certain principles and norms for the transformational processes of the anthropological characteristics

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important for football (Gardasevic, Bjelica, & Vasiljevic, 2020; Gardasevic, & Bjelica, 2019) with morphological characteristics and body composition among them as expected. Findings regarding morphological characteristics and body composition are of crucial importance for complex sports games such as football (Masanovic, T. Bavcevic, & I. Bavcevic, 2019). The morphological space is defined by the longitudinal dimension of the skeleton, the transversal dimensionality of the skeleton, the mass and volume of the body (Gardasevic, Bjelica, & Vasiljevic, 2017). The purpose of knowing morphological characteristics is to improve skills in many sports (Bjelica, Masanovic, & Krivokapic, 2020). The morphological status of top level athletes is relatively homogeneous, depending on the sport, and it can be defined as a model of athletic achievement. Research on morphological characteristics and body composition among athletes of different sports indicates that athletes of different sports have their own specific characteristics. Muscle mass improves performance in activities that require muscular strength and endurance, but also in those that require enviable aerobic ability.

Today, football is certainly the number one sport in the world for its view and popularity (Masanovic, Corluka, & Milosevic, 2018; Masanovic, 2019), and the same applies to Bosnia and Herzegovina and Kosovo, countries of the Southern Balkans, states ex Yugoslavia. The first club that is at the top of the Bosnia and Herzegovina Premier League and is fighting for trophies almost every year is CSC Zrinjski Mostar. The second club is FC Bresa Peje who is successful in the Kosovo Super League. In the 2016/17 competitive season, they both have achieved a staggering success, CSC Zrinjski Mostar was the champion of Bosnia and Herzegovina and FC Besa Peje was the winner of the Kosovo Cup. Based on these two trophies that they have won at the end of the competition season in their countries, both clubs have acquired the right to play on the international football scene within the framework of UEFA's competitions. It became as interesting for researchers to determine the models of anthropometric characteristics and body composition of the players who play for these clubs as to determine the differences among them.

The goal of this research was to analyze the differences in some morphological characteristics and body composition among football players of CSC Zrinjski Mostar and FC Besa Peje.

Method

Sample of subjects

A sample of the subjects consists of a total of 46 elite-level players who performed in the Bosnia and Herzegovina Premier League and Kosovo Super League. The first one consists of 28 players of CSC Zrinjski Mostar, the average age of 24.36±4.14, champion Bosnia and Herzegovina in season 2016/17, and the second one that consists of 18 players FC Besa Peje, the average age of 21.83±3.17, the winner of Kosovo Cup for that season. The football players were tested immediately after the 2016/17 season ended.

Sample of measures

Anthropometric research has been carried in accordance with the International Biological Program. For the purpose of this study, 7 morphological measures have been taken: body height, body weight, waist circumference, triceps skinfold, biceps skinfold, back skinfold, abdominal skinfold, and 3 variables for assessment body composition: body mass index, fat percentage, muscle mass. Anthropometer, caliper, and measuring tape were used for morphological measurements. To evaluate the body composition, Tanita body fat scale - model BC-418MA, was used.

Method of data processing

The data obtained through the research are processed by descriptive and comparative statistical procedures. For each variable, central and dispersion parameters, as well as asymmetry and flattening measures are processed. Differences in morphological characteristics and the composition of the body of the players of these two clubs were determined by using a discriminatory parametric procedure with t-test for small independent samples, with statistical significance of $p<0.05$.

Results

In tables 1 and 2, basic descriptive statistical parameters of morphological characteristics and body composition of the players of the two clubs, where the values of central measurements and dispersion tendencies are calculated, are shown: Arithmetic mean, Standard deviation, Variance, Minimal i Maximal values, coefficient of Curvature and Elongation. First were analyzed the players of CSC Zrinjski Mostar (Table 1).

Table 1. Central and dispersion parameters of variables for assessment of morphological characteristics and body composition of players of CSC Zrinjski Mostar (N=28)

Variables	Min	Max	Mean	S.D.	Variance	Skewness	Kurtosis
body height	170.8	193.0	182.59	4.82	23.27	-.07	.25
body weight	70.0	90.5	78.85	5.80	33.68	.13	-.92
waist circumference	77.0	98.0	86.39	4.35	18.91	.34	.95
triceps skinfold	4.6	13.0	7.59	2.09	4.39	.88	.63
biceps skinfold	3.3	6.2	4.33	.74	.55	1.07	.59
back skinfold	3.7	13.8	9.23	2.18	4.74	.31	.92
abdominal skinfold	4.0	15.0	8.02	2.77	7.66	.89	.36
body mass index	21.4	26.1	23.63	1.14	1.30	.16	-.47
fat percentage	3.9	14.6	8.79	3.18	10.14	-.05	-.88
muscle mass	35.5	46.9	40.67	2.67	7.12	.05	-.03

Legend: Min - Minimal value; Max - Maximal value; Mean - Arithmetic mean; S.D. - Standard deviation; Skewness - coefficient of Curvature; Kurtosis - Elongation

Table 2. shows the analyzed parameters in football players of FC Besa Peje (N=18)

Based on the central and dispersion parameters, the values of the skewness and the kurtosis of all players in two tables, it can be noted that all the variables are placed within the normal distribution boundaries. It can be seen based on the value of skewness as well, that the one variable biceps skinfold of players of both clubs hase mild asymmetry, and though not statistically significant on behalf of better results, it is a positive sign, since it is essential for football players to have lower values of subcutaneous fat tissue and bone mass value. By the value of the kurtosis, it can be seen that the two variables of players of FC Besa Peje - biceps skinfold and abdominal skinfold have a mild leptokurticity, not statistically significant, and two variables body weight and waist circumference have sigificant leptokurticity, which indicates that a greater number of results in this variables are arranged around the arithmetic mean. Generally, according to all sta-

Table 2 Central and dispersion parameters of variables for assessment of morphological characteristics and body composition of players of FC Besa Peje (N=18)

Variables	Min	Max	Mean	S.D.	Variance	Skewness	Kurtosis
body height	169.0	193.0	179.57	6.73	45.36	.40	-.19
body weight	57.9	92.3	73.73	7.11	50.57	.38	2.65
waist circumference	75.0	89.0	80.72	4.35	18.92	.61	-.52
triceps skinfold	4.2	13.0	7.48	2.35	5.52	.72	.14
biceps skinfold	2.8	9.6	4.79	1.88	3.54	1.28	1.04
back skinfold	4.6	10.6	8.36	1.58	2.50	-.97	.64
abdominal skinfold	4.4	18.6	9.35	3.60	12.98	.94	1.14
body mass index	20.0	26.1	22.86	1.54	2.39	.21	.18
fat percentage	4.1	12.6	9.20	2.56	6.55	-.81	-.26
muscle mass	30.4	47.1	37.86	3.33	11.09	.62	3.67

tistical parameters, it can be concluded that here we have some top football players and that the results that prevail are superior to the arithmetic mean, which is not statistically significant for most variables because it is to be expected that regarding players of a professional football club, there is no too large a span between the results of analyzed variables. It can also be stated that the players of FC Besa Peje are significant younger on average, have less body weight than the players of FC Besa Peje, and have a higher fat percentage, though insignificantly. However, a comparative statistical procedure, t-test (Table 3), will show whether it is statistically significant.

Based on the obtained values of t-test results, it can be noted that there are statistically significant differences in three variables at p≤0.01. It is two morphological measures of the body weight and the waist circumference, and it is one of the variables that evaluate the body composition, muscle mass. It can be stated that the football players of CSC Zrinjski Mostar have statistically significantly higher value for all three variables than the players of FC Besa Peje. In all other variables the differences are negligible and not statistically significant.

The significant differences of body weight, waist circumference and muscle mass among the football players of these two clubs are shown in Figure 1.

Table 3. T-test values between the arithmetic means of variables of players of CSC Zrinjski Mostar (N=28) and FC Besa Peje (N=18)

Variables	Football clubs	Mean±SD	Mean Difference	t-test	Sig.
body height	CSC Zrinjski FC Besa Peje	182.59±4.82 179.57±6.73	3.02	1.77	.083
body weight	CSC Zrinjski FC Besa Peje	78.85±5.80 73.73±7.11	5.12	2.674	.010*
waist circumference	CSC Zrinjski FC Besa Peje	86.39±4.35 80.72±4.35	5.67	4.316	.000*
triceps skinfold	CSC Zrinjski FC Besa Peje	7.59±2.09 7.48±2.35	.11	.168	.867
biceps skinfold	CSC Zrinjski FC Besa Peje	4.33±.74 4.79±1.88	-.46	-1.181	.244
back skinfold	CSC Zrinjski FC Besa Peje	9.23±2.18 8.36±1.58	.87	1.465	.150
abdominal skinfold	CSC Zrinjski FC Besa Peje	8.02±2.77 9.35±3.60	-1.33	-1.414	.164
body mass index	CSC Zrinjski FC Besa Peje	23.63±1.14 22.86±1.54	.77	1.932	.060
fat percentage	CSC Zrinjski FC Besa Peje	8.79±3.18 9.20±2.56	-.41	-.463	.645
muscle mass	CSC Zrinjski FC Besa Peje	40.67±3.33 37.86±2.67	2.81	3.164	.003*

Legend: Mean - Arithmetic mean; S.D. - Standard deviation; Sig – significant difference; * - p≤.001

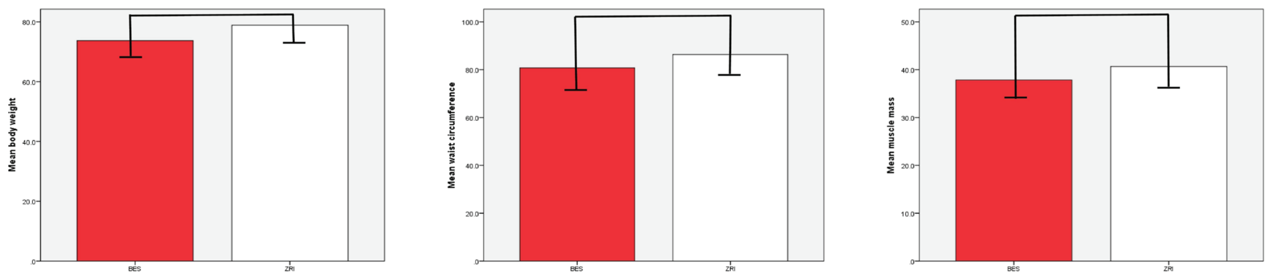

FIGURE 1. The significant differences for three variables among the football players of CSC Zrinjski and FC Besa Peje

Discussion

The goal of this study was to determine the difference in the morphological characteristics and body composition of the top players of the two football clubs, CSC Zrinjski Mostar winner of the Championship Bosnia and Herzegovina and FC Besa Peje, winner of the Kosovo Cup in the 2016/17 season. The results were obtained by using a battery of 11 tests in the area of morphological characteristics and body composition. By looking into the basic descriptive statistical parameters, it can be concluded that we have examined professional sportsmen indeed. It can be noticed that the players of both clubs are of the approximately similar mean values for most variables analyzed, which is not surprising because these are the top two clubs in Bosnia and Herzegovina and Kosovo, and theay have also a similar concentration of good players. The t-test results showed a statistically significant difference in three variables. The first one is one of those that are important for body composition is muscle mass of football players, which has shown that the players of CSC Zrinjski Mostar have significantly higher values than the players of FC Besa Peje. The second variable in which a statistically significant difference has been found is a variable that estimates body weight, where the players of CSC Zrinjski Mostar also have a statistically higher value than the players of FC Besa Peje. Also, at the variable of waist circumference, players of CSC Zrinjski Mostar have shown statistically higher value than the players of FC Besa Peje.

Similar results have been obtained in a recent researches for football players of Montenegrin clubs (Gardasevic, Bjelica, & Vasiljevic, 2019; Bjelica, Gardasevic, & Vasiljevic, 2018; Gardasevic, Bjelica, Popovic, Vasiljevic, & Milosevic, 2018) for football players of other Bosnia and Herzegovinian clubs (Bjelica, Gardasevic, Vasiljevic, Jeleskovic, & Covic, 2019) for football players of other Kosovian clubs (Bjelica, Gardasevic, Masanovic, & Vasiljevic, 2020; Gardasevic, & Bjelica, 2020; Gardasevic, Bjelica, Vasiljevic, Sermaxhaj, & Arifi, 2018). Also, researchers who compared football players of clubs from different countries in the southern Balkans, Montenegrin and Kosovian clubs (Gardasevic, Bjelica, Vasiljevic, & Corluka, 2020; Bjelica, Gardasevic, Vasiljevic, Arifi, & Sermaxhaj, 2019; Gardasevic, Bjelica, Vasiljevic, Arifi, & Sermaxhaj, 2019a; Gardasevic, Bjelica, Vasiljevic, Arifi, & Sermaxhaj, 2019b), Bosnia and Herzegovinian and Montenegrin clubs (Gardasevic, Bjelica, Vasiljevic, & Corluka, 2019; Corluka, Bjelica, Gardasevic, & Vasiljevic, 2019), Bosnia and Herzegovinian and Kosovian clubs (Gardasevic et al., 2020), obtained similar results.

For other variables, some values are better for players of CSC Zrinjski Mostar and some for players of FC Besa Peje, although, insignificantly for statistics, which indicates that these players have very similar morphological characteristics and body composition, which is again, not surprising, considering that these two clubs shared the two trophies in the 2016/17 competitive season in their countries. These results show that ex Yugoslavia had a unique football school, had a football player of similar quality, and that this situation has not changed today in these countries after the breakup of the great Yugoslavia. The values obtained in this research can be useful for coaches of these clubs for making a comparison of their players with others and formulate their work in a way that enables reduction of those parameters that are not good, and raise those that are good to a higher level. That will surely make their football players even better and more successful. Also, both clubs should turn to other researches and check the functional-motoric status, psychological preparation as well as tactical training of their players and analyze whether there is room for their improvement. The results obtained in this research can serve as model parameters for the estimated variables for players of all other football clubs in Bosnia and Herzegovina and Kosovo, because the players that have been analyzed here, were among the best and the most successful football players in Bosnia and Herzegovina and Kosovo at the end of the competitive season in their countries 2016 / 17.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Differences in Some Morphological Characteristics between Students of Younger School Age from Niksic and Mojkovac

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Abstract

Kinesiologists have long ago noticed the great importance of determining and comparing parameters of growth and development in certain age of school children and youth. The aim of the research is to determine eventual existence of statistically significant differences in morphological characteristics between students younger school age from Niksic (Central region of Montenegro) and Mojkovac (Northern region of Montenegro). The research has been performed on a sample of 101 students divided into two groups: 60 students from the second and and third grade of the Primary school "Jagos Kontic" from Niksic and 41 students from the second and and third grade of the Primary school "Aleksa Djilas Beco" from Mojkovac. Morphological characteristics were evaluated by a battery of four variables: body height, body weight, waist circumference and hip circumference. The significance of the differences between students from Niksic and Mojkovac was determinated by t-test for small independent samples. Statistically significant differences in favor of students from Mojkovac have been established in certain morphological parameters. The obtained results of research confirm the need to perform a more comprehensive research based on the comparision of the morphological indicators of students from different parts of Montenegro.

Keywords: *Morphological Characteristics, Differences, Younger School Age*

Uvod

Faktori koji utiču na rast i razvoj jedinke mogu se podijeliti na endogene i egzogene. Od endogenih faktora najvažniji su genetski faktor, odnosno dispozicija, zatim pol, rasa, itd. Veliki uticaj imaju i spoljašnji egzogeni faktori kao što su: uslovi života, tjelesna aktivnost, godišnje doba, higijena i način ishrane (Jovović, 2007). Rast i razvoj djece zauzima značajno mesto u proučavanju cjelokupnog antropološkog statusa djece, kako sa gledišta biološke antropologije, tako i sa gledišta kineziologije (Bala, 2009). Da bi novorođenče dostiglo proporcije odraslog čovjeka, neki djelovi tijela moraju da rastu brže nego drugi tokom postnatalnog razvoja (Hejvud i Gečel, 2017). Postoje izvjesne zakonitosti rasta djeteta koje je bitno poznavati, a to su da intenzivnost rasta pojedinih organa nije uvijek jednaka, trend rasta nije linearan i organi u toku rasta ne povećavaju samo svoju masu, nego mijenjaju i svoju strukturu (Medved, Heimer, Kesić i Pavišić-Medved, 1987). Intenzivnost rasta je najveća u intrauterinom razdoblju života, a zatim postepeno opada. Rast djeteta u visinu i porast tjelesne mase kao najmarkantnijih pokazatelja fizičkog rasta ne idu uporedo. Rast u visinu najviše se odvija na račun rasta koštanog tkiva (noge i kičma), dok je rast u širinu i porast tjelesne mase posledica dijelom rasta koštanog tkiva, dijelom razvoja nervnog, respiratornog, mišićnog i drugih funkcionalnih sistema organizma (Popović, 2008).

Praćenje antropometrijskih promjena u populaciji može biti ključno u sprečavanju buducih problema javnog zdravlja (Popovic, Bjelica, Vukotic i Masanovic, 2018). Mjerenja pojedinih antropometrijskih pokazatelja, počevši od dijagnostičke evaluacije

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(inicijalnog praćenja), doprinose detekciji abnormalnosti rasta i razvoja, ranoj identifikaciji gojaznosti i pothranjenosti. Ona omogućavaju individualniji pristup učenicima i kvalitetnije programiranje nastave fizičkog vaspitanja (Smajić i sar., 2017). Ako se ne poznaju mogućnosti učenika i ako se nastava ne sprovodi po načelima koja proizilaze iz specifičnosti dječijeg uzrasta, mogu nastati veliki problemi, ne samo u vaspitnom i obrazovnom pogledu, nego i u štetnom uticaju na zdravlje i rast i razvoj učenika (Zrnzević, 2006).

U cilju praćenja rasta i razvoja djece, nije korektno koristiti rezultate brojnih istraživanja na odraslim osobama, jer se odavno zna da deca imaju drugačiji rast i razvoj nego adolescenti, tj. mlade osobe (Bala, 2009). U novije vrijeme se nekoliko studija bavilo sagledavanjem morfoloških pokazatelja djece mlađeg školskog uzrasta (Božić-Krstić, Rakić i Pavlica, 2003; Zrnzević, 2006; Zrnzević, 2007; Bigović, Obradović i Krsmanović, 2007; Gligorijević, 2008; Popović, 2008; Gajević i sar., 2010; Vasić i sar., 2011; Ješić, 2017; Smajić i sar., 2017; Malović, 2019). U ovim studijama se osim utvrđivanja nivoa morfoloških pokazatelja, vršilo i njihovo upoređivanje u odnosu na pol, razred, školu ili mjesto boravka učenika.

Morfološki prostor definišu longitudinalna dimenzionalnost skeleta, transverzalna dimenzionalnost skeleta, masa i volumen tijela (Bjelica i Fratrić, 2011). Imajući u vidu komparaciju rezultata pojedinih studija, međusobno i u odnosu na definisane normative, utvrđeno je da postoje izvjesne morfološke diferencije u odnosu na pojedina etnografska područja. Neobičajena visina crnogorskih gorštaka bila je činjenica koju su evropski antropolozi prepoznali prije više od 100 godina (Bjelica, Popovic, Kezunovic, Petkovic, Jurak i Grasgruber, 2012). Takođe, autori su pretpostavljali da je populacija koja živi u Sjevernoj regiji, viša od stanovništva iz Centralne i Južne regije Crne Gore (Milašinović, Gardašević i Bjelica, 2017). Imajući u vidu i mogućnost poređenja mlađe školske populacije iz pomenutih područja, postavljen je cilj ovog istraživanja, koji podrazumijeva utvrđivanje stanja i definisanje razlika u morfološkim karakteristikama učenika mlađeg školskog uzrasta sa teritorija opština Nikšić i Mojkovac. Ostvarivanjem postavljenog cilja, ukazaće se na eventualno postojanje razlika u morfološkim pokazateljima učenika sa dva različita geografska područja u Crnoj Gori, donijeće se određeni zaključci i provjeriti opravdanost daljeg sprovođenja istraživanja vezanih za pomenutu populaciju.

Metod

U pogledu vremenske određenosti istraživanje je transverzalnog karaktera, a sastoji se u jednokratnom mjerenju odgovarajućih morfoloških karakteristika učenika mlađeg školskog uzrasta.

Uzorak ispitanika sačinjavali su učenici uzrasta 8-9 godina koji žive i pohađaju osnovnu školu u Centralnoj (Nikšiću) i Sjevernoj (Mojkovcu) regiji Crne Gore. Ukupan broj ispitanika je 101. Svi ispitanici su podijeljeni na dva subuzorka. Prvi subuzorak je činilo 60 učenika drugih i trećih razreda OŠ "Jagoš Kontić" iz Nikšića, dok je drugi subuzorak činio 41 učenik drugih i trećih razreda OŠ "Aleksa Đilas Bećo" iz Mojkovca.

Mjerenje je sprovedeno u fiskulturnim salama OŠ "Jagoš Kontić" i OŠ „Aleksa Đilas Bećo", a djeca su bila odjevena u sportskoj opremi predviđenoj za realizaciju časa fizičkog vaspitanja.

Uzorak mjera su činili sljedeći morfološki pokazatelji: tjelesna visina, tjelesna masa, obim struka i obim kukova. Tjelesna visina je mjerena antropometrom. Ispitanici su stajali bosi na ravnoj podlozi, u uspravnom stavu, relaksiranih ramena i sastavljenih peta. Glava im je bila namještena tako da je frankfurtska horizontala bila u vodoravnom položaju. Tjelesna masa je mjerena medicinskom vagom, pri čemu su ispitanici stajali bosi na njoj mirno u uspravnom stavu. Obim struka i obim kukova su mjereni neelastičnom santimetarskom trakom, za prvu mjeru na sredini rastojanja karličnog vrha i grudnog koša, odnosno, za drugu mjeru iznad linije koja razdvaja butinu od sedalne regije na tački gdje je maksimalan obim preko stražnjeg dijela. Antropometrijsko mjerenje je sprovedeno uz poštovanje osnovnih pravila i principa vezanih za izbor mjernih instrumenata i tehnike mjerenja, koje preporučuje Inrernacionalni biološki program (IBP).

Dobijeni rezultati su najprije uređeni, a zatim statistički obrađeni na personalnom računaru pomoću softverskog statističkog paketa SPSS 20.0. Podaci su obrađeni postupcima deskriptivne i komparativne statističke procedure. Izračunati su centralni i disperzioni parametri varijabli, a za utvrđivanje razlika u morfološkim karakteristikama između učenika OŠ "Jagoš Kontić" i OŠ „Aleksa Đilas Bećo", primijenjen je t-test za male nezavisne uzorke, sa statističkom značajnošću od $p<0.05$.

Rezultati

U tabelama 1 i 2 prikazani su osnovni statistički deskriptivni parametri morfoloških varijabli učenika OŠ "Jagoš Kontić" iz Nikšića i učenika OŠ "Aleksa Đilas Bećo" iz Mojkovca. Izračunate su sljedeće mjere centralne i disperzione tendencije: aritmetička sredina (Mean), standardna devijacija (S.D.), varijansa (Variance), minimalne (Min) i maksimalne (Max) vrijednosti, koeficijenti nagnutosti (Skewness) i zakrivljenosti (Kurtosis).

Pregledom dobijenih rezultata deskriptivne statistike za subuzorak učenika OŠ "Jagoš Kontić" iz Nikšića, prikazanih u Tabeli 1, konstatuje se sljedeće: vrijednosti skjunisa koje su sa pozitivnim predznakom, ukazuju da su rezultati za većinu ispitanika u zonama slabijih vrijednosti sa normalnom ili umjerenom asimetrijom; negativna vrijednost kurtozisa za varijablu visina tijela ukazuje na platikurtičnost i heterogenost rezultata, dok njegova pozitivna vrijednost za ostale tri varijable koja je iznad 3, ukazuje na leptokurtičnost koju karakteriše izrazita homogenost rezultata.

Tabela 1. Centralni i disperzioni parametri morfoloških varijabli učenika OŠ "Jagoš Kontić"

variable	Min	Max	Mean	SD	Variance	Skewness		Kurtosis	
						Stat	SE	Stat	SE
Tjelesna visina	117.5	148.7	131.9	6.86	47.11	.097	.309	-.110	.608
Tjelesna masa	20.3	54.2	28.1	6.33	40.15	1.77	.309	4.24	.608
Obim struka	50.0	89.8	59.3	7.94	63.12	1.77	.309	3.77	.608
Obim kukova	51.2	88.0	60.6	8.81	46.43	1.65	.309	3.83	.608

Legenda: R – razred; Min. – minimalan rezultat; Max. – maksimalan rezultat; Mean – aritmetička sredina; SD – standarda devijacija; Variance – varijansa; Skewness – skjunis; Kurtosis – kurtozis.

Daljom inspekcijom rezultata u Tabeli 2, koji se odnose na učenike OŠ "Aleksa Đilas Bećo" iz Mojkovca, na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa, konstatuje se da su sve varijable u granicama normalne raspodjele. Vrijednosti skjunisa za varijablu tjelesna visina ukazuju na negativnu asimetriju sa dominacijom boljih rezultata, a za preostale varijable rezultati su za većinu ispitanika u zonama slabijih vrijednosti sa normalnom ili umjerenom asimetrijom. Negativna vrijednosti kurtozisa za varijablu obim kukova ukazuje na platikurtičnost i heterogenost rezultata, dok se u ostalim slučajevima konstatuje leptokuričnost i homogenost dobijenih vrijednosti.

Tabela 2. Centralni i disperzioni parametri morfoloških varijabli učenika OŠ "Aleksa Đilas Bećo"

variable	Min	Max	Mean	SD	Variance	Skewness		Kurtosis	
						Stat	SE	Stat	SE
Tjelesna visina	114.0	155.0	137.0	8.64	74.67	-.489	.369	.761	.724
Tjelesna masa	19.50	58.30	33.6	7.46	55.71	1.01	.369	1.85	.724
Obim struka	48.5	92.0	62.3	8.99	80.99	1.12	.369	1.79	.724
Obim kukova	58.2	85.0	70.7	6.20	38.53	.286	.369	-.371	.724

Legenda: R – razred; Min. – minimalan rezultat; Max. – maksimalan rezultat; Mean – aritmetička sredina; SD – standarda devijacija; Variance – varijansa; Skewness – skjunis; Kurtosis – kurtozis.

Rezultati t-testa za morfološke pokazatelje učenika OŠ "Jagoš Kontić" i učenika OŠ "Aleksa Đilas Bećo", prikazanih u Tabeli 3, ukazali su na statistički značajne razlike u tjelesnoj visini, tjelesnoj masi i obimu kukova. Upoređivanjem srednjih numeričkih vrijednosti navedenih varijabli, uviđaju se značajne razlike u korist učenika iz Mojkovca za varijable: tjelesna visina (5.1 cm), tjelesna masa (5.5 kg) i obim kukova (10.1 cm).

Tabela 3. *Vrijednosti T-testa izmedju aritmetičkih sredina varijabli za procjenu morfoloških karakteristika učenika OŠ "Jagoš Kontić" i učenika OŠ"Aleksa Đilas Bećo"*

Varijable	Škola	Mean	t	df	Sig	MD	SED
Tjelesna visina	J. Kontić	131.9	-3.30	99	.001	-5.11	1.54
	A. Đilas	137.0					
Tjelesna masa	J. Kontić	28.1	-3.91	99	.000	-5.40	1.38
	A. Đilas	33.6					
Obim struka	J. Kontić	59.3	-1.77	99	.080	-3.00	1.69
	A. Đilas	62.3					
Obim kukova	J. Kontić	60.6	-7.60	99	.000	-10.13	1.33
	A. Đilas	70.7					

Legenda: Mean – aritmetička sredina morfoloških varijabli učenika OŠ "Jagoš Kontić i učenika OŠ "Aleksa Đilas Bećo"; t – t vrijednost; df – stepeni slobode; Sig – signifikantnost; MD – razlike aritmetičkih sredina; SED – standardna greška razlike.

Diskusija

Rezultati dobijeni u ovoj studiji su upoređivani sa rezultatima istraživanja iz okruženja za mlađi školski uzrast, kao i sa dostupnim standardima za pojedine parametre. Na osnovu prikazanih rezultata, deskriptivnih statističkih pokazatelja za subuzorak učenika OŠ „Jagoš Kontić" iz Nikšića, uviđa se da je prosječna vrijednost visine tijela 131.9 cm. Prema podacima Nacionalnog centra za zdravstvenu statistiku (Hejvud i Gečel, 2017), za navedeni uzrast se konstatuje raspon vrijednosti tjelesne visine 117-145 cm. Zanimljivo je istaći da se minimalne vrijednosti visine tijela ispitanika obuhvaćenih njihovom studijom, gotovo podudaraju sa minimalnom visinom ispitanika obuhvaćenih ovom studijom (117.5 cm), dok maksimalne vrijednosti visine tijela učenika obuhvaćenih ovom studijom (148.7 cm) premašuju gornju granicu visine tijela ispitanika njihove studije za 3.7 cm. Napredovanje fenomena akceleracije u tjelesnoj visini djece mlađeg školskog uzrasta, može se utvrditi poređenjem rezultata ovog istraživanja sa Normativima antropometrijskih mjera djece i omladine u SFRJ iz 1962. godine (Medved i sar., 1979). Naime, prosječna vrijednost za visinu učenika od 8-9 godina iznosila je 128.1 cm, što je 3.8 cm manje od dobijenih vrijednosti u ovoj studiji. Ova razlika je u potpunosti u saglasju sa konstatacijom antropologa, da je prethodni vijek donio bitno ubrzanje rasta u populaciji industrijski razvijenih zemalja-prosjek 1-1.2 cm za jednu deceniju. Ukoliko dobijene vrijednosti tjelesne mase uporedimo sa normativima za procjenu fizičkog razvoja učenika 7-19 godina u SFRJ (Ivanić, 1988), konstatuje se mala tjelesna masa za dobijenu srednju vrijednost tjelesne visine i mlađi školski uzrast. Dobijeni rezultati za morfološke pokazatelje obim struka i obim kukova su upoređeni sa vrijednostima dobijenim u studiji antropometrijske analize učenika u ruralnom području (Vasić i sar., 2011). I pored izvjesnih sličnosti ruralnog i prigradskog područja, ovdje su se ipak javile određene razlike, koje se svakako mogu pripisati kompleksu dejstva raznovrsnih faktora i specifika genotipskih i fenotipskih modaliteta karakterističnih za različita etnografska područja.

Prosječna vrijednost visine tijela (137 cm) dobijena za subuzorak učenika iz Mojkovca, premašuje za 5.75 cm srednju vrijednost visine tijela učenika istog uzrasta, koju prema podacima Nacionalnog centra za zdravstvenu statistiku navode Hejvud i Gečel (2017). Njihova prosječna visina najsličnija je sa visinom tijela (137.6 cm) djece iz ruralnog područja Bosne i Hercegovine (Vasić i sar., 2011). Obzirom na dobijene rezultate kod učenika iz Mojkovca, koji su više kocentrisani u dijelu boljih od prosjeka, konstatuje se intenzifikacija procesa tjelesnog sazrijevanja za veći dio subuzorka. Tjelesna masa koja pretežno zavisi od faktora voluminoznosti i dimenzionalnosti skeleta, kao i od faktora potkožnog masnog tkiva, kod učenika iz Mojkovca ima podudarne vrijednosti (33.6-33.5 kg) u odnosu na studiju (Vasić i sar., 2011). Prema Normativima za procjenu fizičkog razvoja učenika 7-19 godina u SFRJ (Ivanić, 1988), a na osnovu utvrđene tjelesne visine, učenike iz Mojkovca bi svrstali u grupu sa prosječnom tjelesnom masom. Upoređujući dobijene srednje vrijednosti morfoloških mjera obim struka i obim kukova sa odgovarajućim parametrima učenika iz studije (Vasić i sar., 2011), utvrđena je približna podudarnost. Ovo poređenje, uključujući i prethodna, doprinosi saznanju o veoma značajnom uticaju pojedinih egzogenih faktora (podneblje, nadmorska visina, klimatski faktori, socio-ekonomski uslovi, proizvodni odnosi itd), koji su za ispitanike obije studije izuzetno slični.

Primjenom t-testa, utvrđeno je da između subuzoraka ispitanika postoji statistički značajna razlika u pojedinim morfološkim pokazateljima. Dobijene razlike su u korist učenika iz Mojkovca u morfološkim pokazateljima: tjelesna visina, tjelesna masa i obim kukova. Razlika je najveća u varijabli obim kukova, što i ne čudi, ukoliko se uzmu u obzir razlike u ostalim parametrima longitudinalne i transverzalne dimenzionalnosti. Zapravo, to je pokazatelj prirasta pojedinih dimenzija jednog od funkcionalnih sistema, što je odlika zdravog i naprednog organizma. Dakle, i pored harmonične akceleracije u ovom uzrasnom periodu, moguće je da usljed dejstva određenih faktora ipak dođe do različite dinamike procesa rasta i razvoja, što se može manifestovati i razlikama u pojedinim morfološkim parametrima u odnosu na određena etnografska područja. Takva odstupanja u ovoj studiji, prepoznaju se na osnovu detektovanih diferencija između učenika centralne i sjeverne regije Crne Gore. Navedene razlike su rezultat djelovanja genotipskih, kulturoloških, socio-psiholoških, mikrosocijalnih i mikroklimatskih faktora, a takođe u obzir treba uzeti i fiziološke promjene u organizmu vezane za intenzivan period rasta i sazrijevanja koji će uslijediti.

U skladu sa dobijenim rezultatima, moguće je zaključiti da postoje razlike između subuzoraka ispitanika na osnovu mjerenih morfoloških pokazatelja i pored činjenice da se radi o uzrasnom periodu harmonije i proporcionalnosti razvoja unutrašnjih organa i njihovih funkcionalnih sposobnosti sa razvojem morfoloških karakteristika. Rezultati ovog istraživanja predstavljaju mali doprinos u pravcu rasvjetljavanja problema rasta i razvoja morfoloških parametara djece mlađeg školskog uzrasta. Ovaj period predstavlja veoma senzitivno razdoblje, zbog čega u nastavnim planovima, neizostavno mjesto treba da zauzmu upravo antropometrijska mjerenja, koja će obezbijediti bitne informacije za adekvatno planiranje i sprovođenje sadržaja fizičkog vaspitanja.

Obzirom na veličinu ispitivanog uzorka, uži sistem obuhvaćenih morfoloških varijabli, mora se priznati limitiranost ove studije, što ipak ne umanjuje njenu vrijednost, jer je zapravo ukazala na veliki značaj praćenja pojedinih parametara rasta i razvoja kod djece mlađeg školskog uzrasta. Značajno bi bilo sprovesti obimniju studiju u kojoj bi se vršile komparacije morfoloških pokazatelja djece iz Sjeverne, Centralne i Južne regije Crne Gore. Na taj način bi se nesumnjivo došlo do relevantnih podataka vezano za navedene diferencije.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Nutritional Status of Young School Children in a Rural Environment in Srem District

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Abstract

This research was primarily focused on examining the nutritional status of boys and girls of the young school age. The research was conducted in the rural environment in Srem district and it was based on the Body Mass Index (BMI). This was a transversal research, and the sample included 71 students (34 boys and 37 girls) from the first to the fourth grade of the two primary schools in villages Budjanovci and Nikinci. Measurements included height and weight, and based on them, Body Mass Index (BMI) was calculated. The results of the study showed that the number of underweight children was 1 (1.4%), with normal weight 52 (73.2%), preobese 11 (15.5%) and there were 7 obese students (9.9%). There are no statistically significant differences in the frequency of obesity in all four grades. Considering that obesity at a young school age has a significant effect on obesity and the development of many illnesses in an adult life, it is important to understand this problem in order to do prevention more effectively.

Keywords: *Nutrition, Rural Environment, Students, Body Mass Index*

Introduction

Anthropological status depend on the socioeconomic conditions of the environment in which an individual or a group lives, social standard, the social status of the students' parents, the cultural level of the environment, the basic residential status of an individual and many other factors. The nutitional status is an indicator of the physical ability and health of an individual and an entire population.

Nutritional disorders are one of the most serious health problems. Insufficient nutrition has multiple negative effects on growth and development. Preobesity, as a social health problem, in the period of the young school age, is a predictor of obesity. It shows the risk of appearance of chronic non-communicable diseases and disabilities in later life (Jagodic-Torbica, Ostojić, & Petrović, 2006). For this reason, it is necessary to pay attention and react in time, in order to neutralize these negative effects.

The research that was based on the assumption, that the students living in different environments (urban and rural) have a different degree of development of morphological dimensions and motor abilities, demonstrates the unevenness in terms of the results (Lukić, 2015). Many scientific papers wanted to determine the differences between children from urban and rural environment, between 7 and 14 year-olds in terms of morphological characteristics, motor abilities, degree of nutrition, physical activity, the influence of environmental factors and eating habits. The research about the nutritional status of young school children in urban areas in Vojvodina based on BMI, showed that there are between 11.6 and 35% of preobese and obese children (Obradovic & Srdic, 2007; Lepes, 2009).

Monitoring the nutritional status of the young school children has multiple benefits, because it shows the process of growth and development of the children, it helps to understand the current state, monitoring the effectiveness of Physical Education program and it is an important prognostic factor of health in later life (Pokos, Lauš, & Badrov, 2014; Masanovic, 2019).

The problem arised from the uneveness of the results obtained so far and the lack the research about this topic in Serbia. The aim of this research is to determine the nutritional status of young school children in the rural environment in Srem district based on the body mas index (BMI).

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Method

Sample of respondents

The research was conducted on children of the young school age from the first to the fourth grade in the two primary schools in the villages in Srem district in the municipality of Ruma: „Nebojsa Jerkovic“ in Budjanovci and „Branko Radicevic“ in Nikinci. The sample include 71 students in both schools: 34 boys and 37 girls. The participants were 11 students of the first (5 boys and 6 girls, 7 years old), 17 of the second (9 boys and 8 girls, 8 years old), 20 of the third (11 boys and 9 girl, 9 years old) and 27 students of the fourth grade (9 boys and 14 girls, 10 years old). All students, including their parents and teachers, were fully informed about the test itself, its purpose and methods. They voluntarily took part in the experiment.

Sample variables

Anthropometric measurements of body height expressed in meters (m) and body mass expressed in kilograms (kg) were used in this study. The measurement was carried out in accordance with the recommendations of International Biological Program (IBP) (Weiner & Lourie, 1969). Nutritional status of young school children was assessed by the dependent variable of the Body Mass Index - BMI, which was calculated as the body mass ratio and body height squared according to formula BMI=BM (kg)/BH(m^2) (WHO, 1997). The classification of subjects was done by BMI cut-offs that are used to assess the prevalence of child obesity and skinniness (Cole, Bellizi, Flegal, & Dietz, 2000).

Methods of data analysis

All statistical procedures were performed using the statistical program IBM SPSS Statistics (Statistical Package for the Social Sciences), version 24.0 for Windows. The analysis of nutrition status was done based on school age (grades) and body mass index (underweight - normal weight – preobese - obese). Descriptive statistical parameters were calculated: measures of central tendency and measures of variability - arithmetic mean, standard deviation, coefficient of variation, standard error of arithmetic mean, range of results, minimum and maximum value of measurement results, skewness and kurtosis. Analysis of variance (ANOVA) was used to determine the significance of the differences between the groups of subjects, in grades, in the applied variables system.

Results

In the Table 1, descriptive statistics of body height, body mass of the students from the first to the fourth grade is displayed. It can be concluded that the average results go within the limits of the expected values. Based on the arithmetic mean (AM) for each class, it can be noticed that the difference between 1st and 2nd, 2nd and 3rd, and 3rd and 4th class vary between 4 and 8cm in body height, 0.81 and 6.64 kg in body mass.

Table 1. Descriptive statistics of body height, body mass and body mass index (BMI) for boys and girls, by grade and total

		body height	body mass
Grade	N	AM±SD	AM±SD
1.	11	1.29±0.10	30.56±7.59
2.	17	1.33±0.08	32.21±6.34
3.	20	1.40±0.08	38.15±8.77
4.	23	1.48±0.07	42.82±6.69
Total	71	1.39±0.10	37.07±8.71

Legend: N-number of respondents; AM - arithmetic mean; SD - standard deviation

In Table 2, the descriptive statistics of body mass index (BMI) and the values of the analysis of variance (ANOVA) of students from the first to the fourth grade is presented. It can be concluded that the average results are within the limits of the expected values. Based on arithmetic means (AM) for each class, it can be noticed that the difference between 1st and 2nd, 2nd and 3rd and 3rd and 4th grades by gender and age is between 0.14 and 1.48.

The analysis of variance (ANOVA) shows that the F-test ratio (F = 1.155) of the relationship between the groups (grades) is low and statistically not significant at the locking level (p = 0.000), as we can see it from the result (p = 0.333) of the level of statistical significance for the univariate f test. It can be concluded that the analyzed groups (grades) do not statistically differ from the results in BMI variables.

Table 2. Descriptive statistics of body mass index for boys and girls, by grade and total and analysis of variance (ANOVA)

		body mass index
Grade	N	AM±SD
1.	11	17.98±2.24
2.	17	18.21±3.62
3.	20	19.22±2.93
4.	23	19.82±3.81
Total	71	18.98±3.34
F=1,115; p=0,333		

Legend: N - number of respondents; AM - arithmetic mean; SD – standard deviation; F - univariate f test; p - statistical significance for the univariate f test

An overview of Table 3 can be noticed that 1 (1.4%) child belongs to the group of underweight. The total sample of normally weight children is 52 (73.2%). The children with normal weight are 14 (82.4%) in the second grade, and 7 (63.6%) in the first grade. In the total sample, there are 11 (15.5%) preobese children. The most preobese children are in the first grade (18.2%), and the least in the second grade (11.8%). There are 7 (9.9%) obese children in the total sample. The most obese children are in the fourth grade (13.0%), and the least in the second (5.9%). There are 18 (25.4%) preobese and obese children in the total sample. The largest number is 7 (30.4%) in the fourth grade and the lowest number is 3 (17.7%) in the second grade.

Table 3. Nutrition status by age and total for the whole sample

N=71	1st grade		2nd grade		3rd grade		4th grade		Total	
	boys and girls		boys and girls		boys and girls		boys and girls		boys and girls	
	n	%	n	%	n	%	n	%	n	%
Underweight	1	9.1	0	0	0	0	0	0	1	1.4
Normal weight	7	63.6	14	82.4	15	75	16	69.6	52	73.2
Preobese	2	18.2	2	11.8	3	15	4	17.4	11	15.5
Obese	1	9.1	1	5.9	2	10	3	13	7	9.8
Total	11	100	17	100	20	100	23	100	71	100

Discussion

The results of this study show that among young school children in Srem district have 15,5% preobese and 9,8% obese children. Recent studies show that children of the young school age in 4.6 to 8.2% of cases are obese, ie. in 11.3 to 32% cases are preobese (Đokić & Stojanović, 2010; Đokić & Međedović, 2013).

This research found that there are differences in the nutritional status of young school children in the rural environment in Srem district with the results of previous similar researches. It has been shown that children from the rural environment by percentage of obesity are getting closer to the children from urban areas and that there is no significant difference in the percentage of overweight and obesity by gender (male students / female students) and school age (grades).

The limitation of this study is the fact that morphological characteristics correlate with a certain spatial affiliation, and with the cultural and historical heritage of the environment. The lack of research is because the sample is relatively small. There should be a more detailed survey of the impact of family, education, socioeconomic status and other factors on the state of nutrition.

The cause of preobesity and obesity is the cumulative effect of negative habits and other influences. It can be concluded that one of the cause of this problem lies in professionalism and organizations who work with children, from pre-school and school children to adolescents, as well as the of teaching of physical education. There is a need to focus on teaching physical education and the physical development of children, the prevention of preobesity and obesity and the correct posture, from pre-school to the young school age. It should be carried out by professors of physical education instead of insufficiently and inadequately trained people. It is certain that professors of physical education know far better the motor status of the child the and dynamics of their physical development.

Suggestions for the future research are the introduction of quality content in teaching physical education (theoretical and practical), aimed at preventing obesity and correcting the postural status of children. Based on a much larger number of respondents, we should get more concrete conclusions in what way we can prevent the occurrence of obesity with children.

In our country, there is an insufficient number of research of this issue, and they need to be actualized. There are no national standards for assessment of nutritional status (growth and development tables) of children in Serbia (Pajić, Gardašević, & Jakovljević, 2016). Such research should become part of the teaching of physical education and analysis at the end of each school year, in order to have an insight into the morphological development and state of postural status of children at the levels in one country.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Lifestyle Identification and Degree of Risk of Type 2 Diabetes

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Abstract

The aim of the study was to analyze the relationship between lifestyle level and the risk of type 2 diabetes. The study included 206 respondents who were not diagnosed with diabetes and who were not elite or amateur athletes. The sample was divided into two sub-samples. The first group consisted of respondents aged 18-29 years (n = 104); the second group consisted of respondents aged 30-44 years (n = 102). The FINDRISC questionnaire was used to collect data for the purpose of assessing the risk of type 2 diabetes. The answers to 8 questions provided information about respondent's age, body mass index, waist circumference, physical activity, fruit and vegetable consumption, high-pressure medication, history of hyperglycemia and history of diabetes in the family. The results of the type 2 diabetes risk level indicated the following values: a low level was present in 55%; mild level in 33%; moderate level in 8%; and a high level in 4% of respondents. The results of the study indicated that quality of life decreases with age and the risk of diabetes increases. Therefore, it is important to familiarize people with how to control their weight, their eating habits and their regular physical activity as a preventive method. It is also important to recognize and apply global animation models to maintain and improve the health status of people.

Keywords: *Glycemia, Diabetes, Healthy Lifestyle Habits, Prevention*

Uvod

Dijabetes je hronična i metabolička bolest, praćena povišenim nivoom šećera u krvi, što s vremenom dovodi do ozbiljnih oštećenja srca, krvnih žila, očiju, bubrega, nerava i slično. Prema potvrđenoj klasifikaciji Svjetske zdravstvene organizacije (World Health Organization, 1999), pored dijabetesa uzrokovanog nedostatkom lučenja inzulina (tip 1), trudničkog dijabetesa i drugih specifičnih tipova uzrokovanih genskim poremećajima, lijekovima i hemikalijama, najčešći dijabetes je tip 2 koji je karakterističan po rezistentnosti tijela na inzulin.

Incidencija i prevalencija šećerne bolesti je u porastu, što se povezuje s lošim prehrambenim navikama, ubrzanim načinom života, nedovoljnim kretanjem, stresom i manjkom brige o zdravlju (Šulevski & Kocijan, 2019). Svaka povećana vrijednost glikemije ukazuje na postojanje nekog tipa dijabetesa, te može biti dobar pokazatelj metaboličkog poremećaja (Vega-López, Venn, & Slavin, 2018). Veći rizik od dijabetesa ili srčanih oboljenja obično je povezan s povišenim indeksom tjelesne mase (Meigs et al., 2006). Također, Astrup (2001) navodi da se dijabetes može pripisati pretilosti i prekomjernoj težini s raspodjelom masnih naslaga na trbuhu. Također, visok krvni pritisak u kombinaciji sa dijabetesom može prouzrokovati mnoge komplikacije, a predstavlja dobar pokazatelj prisutnosti stresa koji se obično povećava sa godinama (Emdin, et al., 2015). Bjelica (2015) navodi da se starenjem smanjuje tolerancija prema glikolizi i osjetljivosti na inzulin.

Bivši vrhunski sportisti, većine sportskih disciplina, uživaju bolje zdravstveno stanje u kasnijim godinama života u uporedbi sa opštom populacijom, a što se posebno vidi kod bivših sportista u sportovima izdržljivosti koji imaju nižu incidenciju koronarne srčane bolesti i dijabetesa melitusa tipa 2 (Kujala et al., 2003). Također, anganžman u profesionanom sportu pogoduje dugotrajnosti

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života, dok potencijalni faktori rizika povezani sa smrtnošću, nisu dovoljno prepoznati, a sportski uspjesi, igračka pozicija i predanost mogu utjecati na to (Bjelica, Popovic, Masanovic, & Gardasevic, 2019). Venditti (2007) navodi sveobuhvatne dokaze da način života koji je usmjeren na zdravu prehranu, fizičku aktivnost, te na kraju i gubitak kilograma, može preventivno uticati na nastanak dijabetesa. Prema Hayes & Kriska (2008), posebna važnost se ovakvim pristupom ogleda na smanjenje kardiovaskularnih faktora rizika. Preventivni programi se ne smiju ciljati samo na osobe pogođene bolešću, nego i na porodice, radna mjesta, škole i zajednice (Bazzano, Serdula, & Liu, 2005). Ipak, broj potencijalnih bolesnika može biti značajno smanjen sa ranim otkrivanjem rizik faktora nastanka dijabetesa tipa 2, pa je stoga cilj ovog istraživanja analizirati odnos nivoa životnog stila sa rizikom nastanka dijabetesa tipa 2.

Metod

Istraživanjem je obuhvaćeno 206 ispitanika, slučajnim odabirom, koji nemaju dijagnostikovan dijabetes, te koji se nisu bavili vrhunskim ili amaterskim sportom. Uzorak je podijeljen na dva subuzorka. Prvu grupu ispitanika su činile osobe starosti 18-29 godina (n=104); a drugu grupu su činile osobe starosti 30-44 godina (n=102). U svrhu ovog istraživanja, ispitanici su dobrovoljno ispunili standardni anketni upitnik (Tabela 1), Finnish Diabetes Risk Score (FINDRISC), za procjenu rizika nastanka dijabetesa tipa 2 (Lindström & Tuomilehto, 2003). Upitnikom su obuhvaćene sljedeće karakteristike uzorka: hronološka dob, indeks tjelesne mase (BMI), obim struka, fizička aktivnost, konzumacija voća i povrća, medikamentozno liječenje visokog krvnog pritiska, istorija hiperglikemije i porodična istorija dijabetesa. Naime, bez laboratorijskih analiza ovim upitnikom je moguće predvidjeti rizik od razvoja dijabetesa tip 2. Rizici su stepenovani kroz 5 nivoa i bodovani na slijedeći način: nizak (manje od 7 bodova); blagi (7-11 bodova); umjereni (12-14 bodova); visok (15-20 bodova) i vrlo visok (više od 20 bodova). Podaci su obrađeni putem izračunavanja frekvencija i procentualnih vrijednosti.

Tabela 1. FINDRISC upitnik za procjenu rizika od nastanka dijabetesa tip 2

Starosna dob
0 bodova: 18-29 godina
0 bodova: 30-44 godine
2 boda: 45-54 godine
3 boda: 55-64 godine
4 boda: više od 64 godine
Indeks tjelesne mase (BMI)
0 bodova - manji od 25 kg/m2
1 bod: 25-30 kg/m2
3 boda: veći od 30 kg/m2
Obim struka ispod rebara
0 bodova: manji od 95 cm (muškarci) i 80 cm (žene)
3 boda: 94-102 cm (muškarci) i 80-88 cm (žene)
4 boda: veći od 102 cm (muškarci) i 99 cm (žene)
Jeste li u pravilu na poslu ili u slobodno vrijeme fizički aktivni najmanje 30 minuta svaki dan
0 boda: da
2 boda: ne baš svaki dan
Koliko često jedete voće i povrće
0 bodova: svaki dan
2 boda: ne baš svaki dan
Jeste li ikada uzimali lijekove za visok krvni pritisak
0 bodova: ne
2 boda: da
Jeste li ikad imali povišen šećer u krvi (sistematski pregled ili drugi laboratorijski pregled tokom neke bolesti i kontrole)
0 bodova: ne
5 boda: da
Da li neko od vaše uže porodice ili rođaka ima dijabetes tip 1 ili tip 2
0 bodova: ne
2 boda: da (roditelji roditelja, braća i sestre roditelja, prvi rođaci i sl.)
5 bodova: da (roditelji, brat i sestra, vlastito dijete)

Rezultati

Uvidom u dobivene rezultate (tabela 2), može se konstatovati da je više od polovine ukupnog broja ispitanika sa niskim stepenom rizika od nastanka dijabetesa tipa 2 (55%). Ipak, nešto niži stepen rizika (77%) je bio prisutan kod grupe I (18-29 godina starosti), dok je ostatak ispitivanih imalo blago i umjereno povišeni stepen rizika. Kod grupe I nije bilo ispitanika čiji su rezultati ukazivali na visok stepen rizika od dijabetesa tipa 2. Povećanje stepena rizika je bilo uočljivo kod grupe II (30-44 godine starosti), gdje je najveći postotak ispitanika (47%) imalo blagi rizik, a 33% nizak stepen rizika. U grupi II je takođe uočeno 13% ispitanika sa umjerenim, te 7% sa visokim stepenom rizika.

Tabela 2. Stepen rizika nastanka dijabetesa tip 2 u odnosu na starosnu dob

Stepen rizika	18-29 god. (grupa I)		30-44 god. (grupa II)		Ukupno	
	n	%	n	%	n	%
Nizak	80	77%	34	33%	114	55%
Blagi	20	19%	48	47%	68	33%
Umjeren	4	4%	13	13%	17	8%
Visok	0	0%	7	7%	7	4%
Vrlo visok	0	0%	0	0%	0	0%

Legenda: n – broj ispitanika; % - procenti

U Tabeli 3 je vidljivo da više od polovine ukupnog uzorka ispitanika (55%) ima povišenu tjelesnu masu (BMI >25 kg/m2), od čega je 13% pretilih (BMI >30 kg/m2). Upoređujući indeks tjelesne mase između dvije tretirane starosne grupe, veće prisustvo pretilih odnosno ispitanika sa povišenom tjelesnom masom je prisutno kod II grupe (70%), za razliku od I grupe (42%). Manje od polovine svih ispitanika (44%) je fizički neaktivno, od čega 59% ispitanika iz II grupe i 33% iz I grupe. Takođe, 53% od ukupnog uzorka ispitanika ima povišen obim struka, dok je 54% ispitanika iz ukupnog uzorka izjavilo prisustvo dijabetesa u porodici. Polovina ispitanika neredovno konzumira voće i povrće. Od ukupnog broja, 96% ispitanika je izjavilo da u svojoj prošlosti nisu imali povišen šećer u krvi. S druge strane, 86% ispitanika je izjavilo da su u prethodnom periodu uzimali lijekove za visoki krvni pritisak.

Tabela 3. Karakteristike uzorka prema odgovorima na FINDRISC upitnik

	18-29 god. (grupa I)		30-44 god. (grupa II)		Ukupno	
Godine starosti	**n**	**%**	**n**	**%**	**n**	**%**
18-29	104	50,5%			104	50,5%
30-44			102	49,5%	102	49,5%
Indeks tjelesne mase (BMI)						
<25 kg/m2	61	58%	31	30%	92	45%
25-30 kg/m2	33	32%	54	53%	87	42%
>30 kg/m2	10	10%	17	17%	27	13%
Obim struka ispod rebara						
<94 (80) cm	69	66%	38	37%	107	47%
94-102 m (80-88 ž) cm	29	29%	50	49%	79	42%
<102 m (88 ž) cm	6	6%	14	13%	20	11%
Tjelesna aktivnost najmanje 30 minuta svaki dan						
Da	69	66%	42	41%	111	56%
Ne	35	33%	60	59%	95	44%
Često konzumiranje voća i povrća						
Da	52	50%	51	50%	103	50%
Ne	52	50%	51	50%	103	50%
Uzimanje lijekova za visok krvni pritisak (u prošlosti)						
Da	92	88%	86	84%	178	86%
Ne	12	12%	16	16%	28	14%
Povišen šećer u krvi (u prošlosti)						
Da	103	99%	95	93%	198	96%
Ne	1	1%	7	7%	8	4%
Dijabetes u porodici						
Ne	51	49%	46	45%	97	46%
Da (Tip 1)	37	36%	34	33%	77	34%
Da (Tip 2)	16	15%	22	22%	46	20%

Legenda: n – broj ispitanika; % - procenti

Diskusija

Poznata je činjenica da se rizik nastanka šećerne bolesti povećava s godinama starosti. Pored nasljednih faktora na koje čovjek svojim izborom ne može uticati, nezdrave životne navike u velikoj mjeri povećavaju rizik od nastanka šećerne bolesti. Prisustvo povećane tjelesne mase i pretilosti kod više od polovine ispitanih može sugerisati na rizik od nastanka dijabetesa tipa 2. Trend povećanja indeksa tjelesne mase (BMI) najčešće prati i trend povećanja obima struka. Ova dva navedena parametra se mogu povezati sa nedostatkom energetske potrošnje. Naime, BMI i obim struka se teško mogu održati prihvatljivim u slučaju fizičke neaktivnosti. Činjenica da se tek nešto više od polovine ispitanika bavi fizičkim aktivnostima, pretpostavlja prisustvo nedostatka kretanja odnosno hipokinezije.

Da je dijabetes rasprostranjena bolijest potvrđuje i činjenica o njenom prisustvu u bližoj porodici kod više od polovine tretiranih ispitanika. Rizik od povišene glikemije se takođe povećava s godinama starosti. Mlađi ispitanici (18-29 godina) u većini slučajeva nisu imali povišen nivo glikemije, za razliku starijih ispitanika (30-44 godine). Takođe, važno je istaknuti poznatu činjenicu da dijabetes tip 2 nasljedna bolijest za razliku od ostalih tipova dijabetesa.

Prethodna istraživanja značajno su osvijetlila napredak u razumijevanju regulacije molekularnih mehanizama te učinaka vježbanja na iskorištavanje glukoze u skeletnim mišićima, pri čemu je ustanovljeno da GLUT4 nosi glavnu ulogu u regulaciji transporta glukoze tokom vježbanja, kao i u jačanju osjetljivosti na inzulin nakon vježbanja (Hayashi, Wojtaszewski, & Goodyear, 1997). Zdrave životne navike, koje su utemeljene na redovnoj fizičkoj aktivnosti i konzumaciji zdravih namirnica, mogu preventivno uticati na pojavu šećerne bolesti. Intenzivnom izmjenom načina života, te promjenama u tjelesnoj težini i fizičkoj aktivnosti, moguće je ostvariti značajno smanjenje incidencije dijabetesa (Ilanne-Parikka & Eriksson, 2008). Takođe je utvrđeno relativno smanjenje progresije dijabetesa od 58% kod osoba sa promjenom ka zdravim životnim navikama (Lindstrom & Louheranta, 2003).

Dnevna fizička aktivnost, od laganog do umjerenog intenziteta i u trajanju od najmanje 30 minuta dnevno, može da smanji rizik od dijabetesa tipa 2, za razliku od neaktivnog stila života. Fizička neaktivnost se procjenjuje glavnim uzrokom za otprilike 27% dijabetesa i otprilike 30% ishemijske bolesti srca (World Health Organization, 2010). Prednosti fizičke aktivnosti ogledaju se u kontroli tjelesne težine i redukciji masnih naslaga, pogotovo onih u trbušnoj regiji. Odgovarajuće količine vježbanja održavaju volumen mišića i povećavaju korištenje glukoze (Shoda, Oh, Shida, & Tanaka, 2015). Suštinski, zdravstvena uloga vježbanja se ogleda u izazivanju adaptivnih, morfoloških i funkcionalnih promjena tijela, što se reflektuje u poboljšanju zdravstvenih pokazatelja (Bjelica, & Krivokapić, 2019).

Preporuka je koristiti raznovrsne fizičke aktivnosti, uz odabir prema ličnim afinitetima. Optimalan izbor ovakvih aktivnosti bi trebao prvenstveno pozitivno uticati na zdravlje, te smanjiti rizik od povrede. Raznovrsne aerobne aktivnosti (hodanje, trčanje, vožnja bicikla, plivanje, nordijsko skijanje i slično) preporučljivo je kombinovati sa vježbama snage (otpor vlastitog tijela, slobodni tegovi, trenažeri i slično). Takođe, činjenica da skoro polovina ispitanika redovno ne konzumira voće i povrće, ukazuje na blago izraženu svijest o važnosti pravilne prehrane. Veći unos voća, posebno bobičastog, zatim zelenog lisnatog i žutog povrća, povezan je s manjim rizikom od dijabetesa tipa 2 (Wang, Fang, Gao, Zhang, & Xie, 2016).

Može se zaključiti, da povećan indeks tjelesne mase i obim struka, nedostatak konzumacije voća i povrća, propraćeno sa smanjenim nivoom fizičke aktivnosti, uzrokuje veći stepen rizika od nastanka dijabetesa tipa 2. Takođe, procesom biološkog starenja se kvaliteta praktikovanja zdravih životnih navika postepeno smanjuje, čime se i rizik od nastanka bolijesti povećava. Zdravstvena politika u saradnji sa sportskim sistemom treba imati značajnu ulogu u prevenciji dijabetesa. Ljude je važno upoznati sa načinom kontrole tjelesne težine, zdravim prehrambenim navikama i upražnjavanjem redovne fizičke aktivnosti, kao preventivnog načina djelovanja. Takođe, važno je prepoznati i primijeniti globalne modele animacije s ciljem održavanja i unapređenja zdravstvenog statusa. Limitiranost ove studije se ogleda u nedostatku ispitanika >45 godina starosti, a posebno ispitanika treće životne dobi (> 65 godina starosti), s obzirom na poznatu činenicu da se radi o najugroženijoj populaciji. Takođe, nedostatkom se smatra nepostojanje informacija o spolu tretiranih ispitanika.

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ORIGINAL SCIENTIFIC PAPER

Attitudes of Teachers towards Physical Education Teaching in Elementary School

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Abstract

Physical education is a very important teaching subject that affects the growth and proper development of children. The main goal of this research is to determine the teachers attitudes to the teaching on Physical Education. The sample of respondents is consisted of 31 teachers from Podgorica and Niksic. Respondents are divided into 2 subsamples according to the criterion of residence: Podgorica and Niksic. It was used a standardized "Secondary PE Teacher Survey Questionnaire" adapted for this research. The results are shown numerically and in percentages, and statistically significant differences are calculated by the Hi square test at the significance level of p = 0.01. The results of the research showed a positive attitude of the teachers towards the teaching of Physical Education, as well as the fact that the teachers are aware of the importance of teaching and conducting the teaching in a well-organized way.

Keywords: *Teachers, Primary school, Attitudes, Physical Education*

Uvod

Fizičko vaspitanje je specifično vaspitno područje koje doprinosi razvoju čovjeka u cjelini, to nije samo podsticanje rasta i razvoja, razvijanje tjelesnih sposobnosti i unaprjeđenje zdravlja, već i znalačko korišćenje sistema fizičkih vježbi, igara i sportova, kojima se utiče na cjelokupni razvoj čovjeka (Šimleša i Potkonjak, 1989). Kroz nastavu Fizičkog vaspitanja, djeca zadovoljavaju svoje osnovne potrebe za motornom aktivnošću, fizičkim vježbanjem, fizičkim razvitkom, intelektualnim razvitkom, zdravljem i socijalizacijom (Bjelica i Krivokapić, 2010, Bjelica i Krivokapić, 2019). Iz navedenog, primjećuje se da se, kroz raznovrsne aktivnosti na času Fizičkog vaspitanja, utiče na psiho-fizički razvoj djeteta, kao i zadovoljavanje njegovih potreba za igrom i kretanjem (Bjelica i Krivokapić, 2011; Bjelica i Krivokapić, 2019).

Kako bi se obezbijedilo sprovođenje kvalitetne nastave Fizičkog vaspitanja, neophodni su dobri materijalni uslovi, kao i stručni nastavnici koji dobro poznaju sadržaj nastave i psiho-fizičke mogućnosti učenika. Brojni autori ističu važnost nastavnika u samom procesu nastave Fizičkog vaspitanja (Ilić i Stević, 1996; Bjekić, 1999; Višnjić, Jovanović, i Miletić, 2004). Prema navedenim autorima, nastavnik mora biti stručno osposobljen za izvođenje nastave, dobro pokazivati i objašnjavati sadržaj, kao i adekvatno ocjenjivati. Takođe, mora pokazivati pozitivan odnos prema Fizičkom vaspitanju, ali i uspostavljati dobre odnose sa djecom, kako bi ih motivisao za rad. Nažalost, opšte je poznato da mnoge škole nemaju adekvatne uslove za izvođenje nastave Fizičkog vaspitanja (N. Zrnzević i J. Zrnzević, 2015), da se nastava Fizičkog vaspitanja često ne sprovodi artikulisano i sistemski i da sadržaj nije prilagođen potrebama djece (Krsmanović, 1996; Zrnzević, 2007). Istraživanja sprovedena u regionu pokazuju da se nastava Fizičkog vaspitanja ne sprovodi redovno, prema rasporedu, što je zabrinjavajuće s obzirom na njegovu važnost za razvoj i zdravlje djeteta (Dobraš, Kukrić, Petrović, i Stojanović, 2016). Na neadekvatan odnos prema nastavi Fizičkog vaspitanja ukazuje i činjenica da je, u istom istraživanju, samo 23% djece navelo Fizičko vaspitanje kao omiljeni predmet (Dobraš et al., 2016). Međutim, brojna istraživanja su ukazala i na pozitivne stavove djece prema Fizičkom vaspitanju (D. Lazarević, Orlić, B. Lazarević, i Ranisa-

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vljević-Jelić, 2015; E. Međedović, Murić, i A. Međedović, 2005; Radisavljević i Višnjić, 2004; Mitrovic & Dragutinovic, 2019), što je dijelom zasluga i nastavnika, kao jednog od nosioca nastavnog procesa (Bjekić, 1999).

Iz navedenog, primjećuje se da je uloga nastavnika veoma važna, pa je neophodno utvrditi kakvi su njihovi stavovi prema nastavi Fizičkog vaspitanja, što i jeste osnovni cilj ovog istraživanja. Na osnovu dobijenih rezultata, moguće je utvrditi koliko nastavnici uvažavaju Fizičko vaspitanje kao nastavni predmet, i preduzeti određene mjere za podsticanje pozitivnih stavova prema ovom nastavnom predmetu, radi poboljšanja kvaliteta nastave.

Metod rada

Ukupan uzorak ispitanika činio je 31 nastavnik razredne nastave, podijeljen na dva subuzorka: 18 nastavnika Osnovne škole "Vuk Karadžić" iz Podgorice i 13 nastavnika Osnovne škole "Luka Simonović" iz Nikšića.

Istraživačka tehnika kojom su utvrđeni stavovi prema Fizičkom vaspitanju bila je anketa, u okviru koje je korišćen standardizovani upitnik "Secondary PE Teacher Survey" koji je modifikovan i prilagođen nastavnicima razredne nastave (North Western Counties Physical Education Association, 2014). Upitinici su se popunjavali u prosjeku 15 minuta, a svi ispitanici su dobrovoljno pristali da učestvuju u ovom istraživanju. Ono što je važno naglasiti, jeste to da je anketa bila anonima, pa je samim tim nastavnicima garantovana stroga povjerljivost odgovora. Podaci su se prikupljali u pomenutim školama, za vrijeme nastave. Nakon sprovedenog istraživanja, uvidom u popunjene upitnike, odmah su iz dalje obrade rezultata isključeni nepotpuni i nejasno popunjeni upitnici. Dobijeni podaci su analizirani uz pomoć softverskog statističkog paketa SPSS 20.0 (Chicago, IL, USA). Rezultati su u tekstu prikazani tabelarno u procentima, a razlike u stavovima prema nastavi Fizičkog vaspitanja između navedenih subuzoraka utvrđene su Pirsonovim H2 (HI kvadrat) testom na nivou značajnosti od $p<0.05$.

Rezultati

Dobijeni rezultati ovog istraživanja prikazani su u tabelama 1 i 2, odvojeno za nastavnike iz Podgorice i nastavnike iz Nikšića. Dalji uvid u njihovu sadržinu daće nam jasniji pregled rezultata.

Tabela 1. Stavovi nastavnika razredne nastave prema Fizičkom vaspitanju - Podgorica

Pitanja	Odgovori	Broj	Procenat (%)
1. Da li smatrate da je Fizičko vaspitanje važno za učenike u bilo kom od razreda u Vašoj školi?	a) Da	19	100
	b) Ne	0	0
2. Mogu li učenici biti izuzeti od obaveznog fizičkog obrazovanja na jedan period ocjenjivanja ili duži za svaki od sledećih razloga?	a) Škole sporta	18	95
	b) Školski sport	5	26
	c) Sportske aktivnosti u zajednici	16	84
	d) Fizička onesposoljenost	0	0
3. Da li se standardi Vaše škole za Fizičko vaspitanje konkretno odnose na sledeći?	a) Kompetentnost	19	100
	b) Razumijevanje	19	100
	c) Redovnost	19	100
	d) Nivo spremnosti	19	100
4. Da li Vaša škola ima pisani program za Fizičko vaspitanje?	a) Da	19	100
	b) Ne	0	0
5. Da li Vaš nastavni program fizičkog vaspitanja uključuje sljedeće komponente?	a) Ciljeve	19	100
	b) Aktivnosti	19	100
	c) Ocjenjivanje	19	100
6. Koji od sljedećih najbolje opisuje tipičan brojčani odnos učenika i nastavnika na časovima Fizičkog vaspitanja u Vašoj školi?	a) do 19 učenika	6	31
	b) do 29 učenika	11	58
	c) preko 30 učenika	2	11
7. Kada pripremate časove za Fizičko vaspitanje, koji procenat vremena obično izdvajate za učenike da budu fizički aktivni?	a) 0 do 24%	0	0
	b) 25 do 49%	4	21
	c) 50 do 74%	2	11
	d) 75 do 100%	13	68
8. Da li u Vašoj školi koriste sljedeću tehnologiju prilikom podučavanja na časovima Fizičkog vaspitanja?	a) Računari	3	16
	b) Video kamere	0	0
	c) DVD	0	0
9. Da li Vaša škola prikuplja informacije o morfološkim karakteristikama učenika?	a) Da	18	95
	b) Ne	1	5
10. Koji od sljedećih kriterijuma koristite za ocjenjivanje učenika iz Fizičkog vaspitanja?	a) Pohađanje	19	100
	b) Oprema	19	100
	c) Učešće	19	100
	d) Testovi	19	100
11. Da li se ocjene za Fizičko vaspitanje smatraju jednakim ocjenama kao kod drugih predmetnih područja pri određivanju prosjeka ocjena?	a) Da	19	100
	b) Ne	0	0

Pregledom Tabele 1, uočavamo da su nastavnici razredne nastave Fizičkog vaspitanja manje-više bili saglasni, te se kod nekoliko pitanja mogu naći identični odgovori svih ispitanika. Iz odgovora na prvo pitanje, vidimo da svi nastavnici smatraju da je Fizičko vaspitanje veoma važan nastavni predmet u bilo kom razredu (100%). Ono oko čega se nisu složili, jesu razlozi zbog kojih treba dopustiti učeniku da izostane sa redovne nastave ovog predmeta. Specifičnost kod ovog pitanja jeste to što su nastavnici pored svakog razloga imali ponuđenu mogućnost da zaokruže da ili ne, odnosno da prihvate taj razlog ili da ga odbace. Gotovo jednoglasno su se složili da škole sporta nisu razlog za odsustvo sa nastave (95%), dok su kod školskog sporta (26%) i sportskih aktivnosti u zajednici (84%) bili podijeljeni. Jednoglasno su se izjasnili da fizički onesposobljenu osobu, treba izuzeti iz nastave. Kada su u pitanju postojanje lične kompetencije nastavnika i standardi koji se poštuju u njihovoj školi kada je u pitanju nastava Fizičko vaspitanje, jednoglasno je potvrđeno da postoje: kompetentnost (100%), razumijevanje (100%), redovnost (100%) i nivo spremnosti (100%). Takođe, potvrdili su da u njihovoj školi postoje pisani programi za ovaj predmet (100%), kao i to da oni sadrže: ciljeve (100%), aktivnosti (100%) i ocjenjivanje (100%). Čest problem, koji remeti efikasno izvođenje nastave jeste prekomjeran broj djece na ograničenom prostoru u kojem se nastava odvija. Najčešće su se izjasnili da je na časovim prisutno do 29 učenika (58%), manje ispitanika dalo je odgovor da je na času prisutno do 19 učenika (31%) a najmanje njih je dalo odgovor da je na času prisutno preko 30 učenika (11%). Njih 68% se izjasnilo da je aktivnost učenika na času veoma visoka, čak od 75 do 100% ukupnog vremena. Primjena tehnologije u nastavi Fizičkog vaspitanja je, prema datim odgovorima, veoma slabo zastupljena. Samo 16% njih koristi računare u nastavi, a ostala sredstva tehnologije uopšte ne. Međutim, redovno sprovode testiranja i prate razvoj učenika, te njih 95% ima prikupljene podatke o morfologiji učenika. Prilikom ocjenjivanja učenika, vrednuju sve stavke koje su bile ponuđene kao ključne u ocjenjivanju: pohađanje nastave (100%), oprema (100%), učešće (100%) i testovi (100%). Te su time pokazali da vrednuju Fizičko vaspitanje kao i sve ostale predmete (100%).

Tabela 2. Stavovi nastavnika razredne nastave prema Fizičkom vaspitanju - Nikšić

Pitanja	Odgovori	Broj	Procenat (%)
1. Da li smatrate da je Fizičko vaspitanje važno za učenike u bilo kom od razreda u Vašoj školi?	a) Da	12	100
	b) Ne	0	0
2. Mogu li učenici biti izuzeti od obaveznog fizičkog obrazovanja na jedan period ocjenjivanja ili duži za svaki od sledećih razloga?	a) Škole sporta	12	100
	b) Školski sport	2	17
	c) Sportske aktivnosti u zajednici	2	17
	d) Fizička onesposoljenost	0	0
3. Da li se standardi Vaše škole za Fizičko vaspitanje konkretno odnose na sledeći?	a) Kompetentnost	12	100
	b) Razumijevanje	12	100
	c) Redovnost	12	100
	d) Nivo spremnosti	12	100
4. Da li Vaša škola ima pisani program za Fizičko vaspitanje?	a) Da	12	100
	b) Ne	0	0
5. Da li Vaš nastavni program fizičkog vaspitanja uključuje sljedeće komponente?	a) Ciljeve	12	100
	b) Aktivnosti	12	100
	c) Ocjenjivanje	12	100
6. Koji od sljedećih najbolje opisuje tipičan brojčani odnos učenika i nastavnika na časovima Fizičkog vaspitanja u Vašoj školi?	a) do 19 učenika	0	0
	b) do 29 učenika	0	0
	c) preko 30 učenika	12	100
7. Kada pripremate časove za Fizičko vaspitanje, koji procenat vremena obično izdvajate za učenike da budu fizički aktivni?	a) 0 do 24%	0	0
	b) 25 do 49%	0	0
	c) 50 do 74%	12	100
	d) 75 do 100%	0	0
8. Da li u Vašoj školi koriste sljedeću tehnologiju prilikom podučavanja na časovima Fizičkog vaspitanja?	a) Računari	0	0
	b) Video kamere	0	0
	c) DVD	0	0
9. Da li Vaša škola prikuplja informacije o morfološkim karakteristikama učenika?	a) Da	8	67
	b) Ne	4	33
10. Koji od sljedećih kriterijuma koristite za ocjenjivanje učenika iz Fizičkog vaspitanja?	a) Pohađanje	12	100
	b) Oprema	12	100
	c) Učešće	12	100
	d) Testovi	12	100
11. Da li se ocjene za Fizičko vaspitanje smatraju jednakim ocjenama kao kod drugih predmetnih područja pri određivanju prosjeka ocjena?	a) Da	12	100
	b) Ne	0	0

Uvidom u Tabelu 2, vidimo da su nastavnici razredne nastave koji rade u Nikšiću bili još saglasniji prilikom popunjavanja ove ankete. Te su tako svi na prvo pitanje, koje se ticalo važnosti Fizičkog vaspitanja, kao nastavnog predmeta u bilo kom uzrastu, jasno iznijeli pozitivan stav (100%). Drugo pitanje je jedino podijelilo nastavnike razredne nastave, pa su i ovdje škole sporta jedan od razloga koje najmanje treba uvažavati kao razloge za izuzeće iz nastave (100%), dok je učešće u školskom sportu i sportskim aktivnostima u zajednici dovoljan razlog da se učeniku odobri izuzeće (17%). Takođe, svi standardi vezani za ovaj nastavni predmet koji su važili i za nastavnike u Podgorici, važe i ovdje: kompetentnost (100%), razumijevanje (100%), redovnost (100%). Pisani program (100%), kao i ciljevi, aktivnosti i ocjenjivanje, zastupljeni su i u ovoj školi (100%). Kako je već navedeno, najčešći problem, koji se javlja pri izvođenju nastave Fizičkog vaspitanja, jeste brojna grupa koja u ovom slučaju uvijek iznosi preko 30 učenika (100%). Opterećenje, tj. aktivnosti učenika u toku časa su uglavnom od 50 do 74% (100%). Ponuđenu tehnologiju u nastavi Fizičkog vaspitanja, niko od anketiranih nastavnika razredne nastave ne koristi. Informacije o morfološkom statusu učenika posjeduje 67% nastavnika. U kriterijume za ocjenjivanje spada sve što je ponuđeno, a bitne stavke ocjenjivanja za ovaj nastavni predmet su: pohađanje (100%), oprema (100%), učešće (100%) i testovi (100%). Primjećuje se takođe, da svi koji su učestvovali u ovom istraživanju vrednuju Fizičko vaspitanje jednako kao i sve ostale predmete (100%).

Tabela 3. Razlika stavova nastavnika razredne nastave iz Podgorice i Nikšića prema Fizičkom vaspitanju kao predmetu utvrđena Hi kvadrat testom

Variables	Pearson Chi-Square	Df	Asymp. Sig. (2-sided)
PITANJE2A	.653	1	.419
PITANJE2B	9.574	1	.002*
PITANJE2C	7.888	1	.005*
PITANJE6	23.774	2	.000*
PITANJE7	4.699	2	.095
PITANJE8A	2.098	1	.148
PITANJE9	4.284	1	.038*

Legenda: * - postoji statistički značajna razlika

Pregledom Tabele 3, primjećuje se da postoji značajna razlika između stavova nastavnika razredne nastave iz Podgorice i Nikšića kada su u pitanju stavovi prema Fizičkom vaspitanju, tj. prema nekim aspektima nastave vezanim za izostajanje učenika, organizaciju nastave, te praćenje razvoja učenika. Statistička značajnost postoji kod 4 varijable. Razlike su se javile kod drugog pitanja gdje su dati razlozi za opravdanost izuzeća učenika sa nastave, gdje su različita mišljenja o tome treba li se učenicim dozvoliti izostajanje ukoliko učestvuju u aktivnostima kao što su školski sport (Sig=.002) i sportske aktivnosti u zajednici (Sig.=.005). Različito mišljenje pokazuju odgovori na šesto pitanje, gdje je najveća statistička razlika (Sig.=.000), gdje su učitelji iz Nikšića jasno sugerisali na problem brojnosti učenika u toku časa a učitelji iz Podgorice nijesu. Na kraju, kod devetog pitanja takođe se uočava razlika, tj. bilo je podijeljenih odgovora o prikupljanju informacija o učenicima (Sig.=.038), i nastavnici iz Podgorice su u većem broju potvrdili postojanje ovakvih podataka.

Diskusija

Poznato je da Fizičko vaspitanje doprinosi normalnom razvoju organizma, jačanju zdravlja, snaženju i čeličenju organizma, osposobljavanje za raznovrsnu pokretljivost, stvaranju higijenskih navika i obezbjeđivanju aktivnog odmora (Bakovljev, 1997). Uzevši u obzir činjenicu da savremeni način života redukuje fizičku aktivnost djece, to još više naglašava važnost Fizičkog vaspitanja kao školskog predmeta (Đorđić, 2005). Takođe, za sprovođenje kvalitetne nastave neophodni su kompetentni nastavnici, koji bi trebalo da podstiču učenike na bavljenje fizičkim aktivnostima (N. Zrnzević i J. Zrnzević, 2015; Dragutinovic & Mitrovic, 2019). Kako bi nastavnici razredne nastave pozitivno djelovali na učenike, neophodno je da i oni sami imaju pozitivne stavove prema nastavi fizičkog vaspitanja. Rezultati dobijeni u ovom istraživanju ukazuju da su nastavnici razredne nastave izuzetno pozitivno orijentisani prema nastavi Fizičkog vaspitanja, poštujući plan i program i pripremajući adekvatan nastavni sadržaj. Što se tiče razlika stavova među nastavnicima razredene nastave iz Podgorice i Nikšića, statistička značajnost javila se kod 3 varijable, i to kada je u pitanju opravdanost izuzeća sa nastave, brojnost učenika na nastavi i prikupljanje podataka o antropometrijskom statusu učenika. Najveći problem svakako predstavlja preveliki broj učenika na času u OŠ "Luka Simonović" u Nikšiću, preko 30, što svakako otežava rad. Rješenje se može potražiti u izmjenama rasporeda korišćenja fiskulturne sale, jer često dolazi do poklapanja termina među odjeljenjima, a samim tim veliki broj učenika utiče na kvalitet nastave.

Dobijeni pozitivni stavovi nastavnika razredne nastave su izuzetno važni, jer pokazuju želju nastavnika da kvalitetno pripreme i sprovedu nastavu Fizičkog vaspitanja. S obzirom na to, nastavnik je, preko određenih pedagoških situacija, u mogućnosti da podstiče i usmjerava učenika na nastavi, kao i da omogući veću uključenost i saradnju tokom nastave (Havelka, 2000; Stojiljković, 2014). Zahvaljujući tome, moguće je doprinijeti većoj aktivnosti učenika, koja je neophodna za njegov psiho-fizički razvoj (Bjelica i Petković, 2009; Masanovic, 2019; Masanovic, 2020).

Rezultati dobijeni u ovom istraživanju korenspodenti su rezultatima prethodnih istraživanja na ovu temu, u kojima nastavnici razredne nastave takođe pokazuju pozitivne stavove prema nastavi Fizičkog vaspitanja (E. Međedović i A. Međedović, 2007). Međutim, i pored toga, neophodno je i dalje raditi na razvijanju pozitivnih stavova, kako nastavnika razredne nastave, tako i učenika prema nastavi Fizičkog vaspitanja, s obzirom na njen značaj za razvoj cjelokupne ličnosti djeteta. Pozitivne stavove potrebno je "prenijeti" i u fiskulturne sale, kako bi djeca shvatila značaj i zaista zavoljela nastavu Fizičkog vaspitanja.

Svrha ovog istraživanja bila je ispitati stavove nastavnika razredne nastave prema Fizičkom vaspitanju, kao nastavnom predmetu, jer najviše od njih zavisi da li će i koliko djeca biti fizički aktivna. Kao neko ko je uz djecu od mlađeg školskog uzrasta, a onda i kroz dalje školovanje, jako je važno da od samog početka motivišu i zainteresuju djecu da se bave sportskim aktivnostima. Složićemo se da to može samo nastavnik koji poštuje svoj predmet, motivisan i dovoljno obučen za rad. Imajući u vidu rezultate koje smo dobili ovim istraživanjem evidentno je da treba poraditi na prostoru i opremi za izvođenje ove nastave, ali ne zaboraviti seminare i obuke, koje je potrebno organizovati u većem broju kako bi se nastavnici dodatno edukovali i pospiješili svoju nastavu, što su oni svojim odgovorima i potvrdili.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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REVIEW PAPER

A Content Analysis of Published Articles in Montenegrin Journal of Sports Science and Medicine from 2019 to 2020

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Abstract

Montenegrin magazine for sports science and medicine "Montenegrin Journal of Sports Science and Medicine" (MJSSM), is the only magazine in Montenegro with the index in two world databases: Web of Science and Scopus. Articles published in this magazine stem from different fields of sports science: biomechanics, physiology, sports medicine, anthropology, methodology and other fields related to sports. In the previous three editions of this magazine, 30 articles have been published which are classified according to the following science fields: Sports Medicine, Sports Training, Anthropology of Sport; Sports Psychology; Sports Physiology; Systematic Analysis of Sports Articles. The analysis revealed that the articles in the field of Sports Training are the most numerous, therefore the most cited works as well as the best positioned. Sports Training is an integral part of a triage technology which enables an athlete's transformation processes of certain segments of the anthropological status as well as an improvement of technical and tactical elements which ultimately lead to the realization of the top sports achievements. Also, results published in the magazine "Montenegrin Journal of Sports Science and Medicine" in the field of Sports training could be beneficial in monitoring athletes, notifying corrections during training methods, estimating and improving capability differences among athletes from different sports orientations.

Keywords: *Science, Journal, Sport, Medicine, Montenegro*

Uvod

Crnogorski časopis za sportsku nauku i medicinu Montenegrin Journal of Sports Science and Medicine (MJSSM) je jedini časopis u Crnoj Gori koji je indeksiran u dvije svjetske baze podataka, Web of Science i Scopus. Časopis se objavljuje dva puta godišnje, u martu i septembru, a posljednje izdanje donosi 10 naučnih radova koja potpisuju poznata imena 38 naučnika iz svih krajeva svijeta. Riječ je o naučnicima iz SAD-a, Turske, Španije, Islanda, Irana, Portugala, Ujedinjenog Kraljevstva i Poljske. Sa naučnog aspekta, pojedine teme koje se obrađuju u časopisu MJSSM su podrška trenerima u praktičnom radu. Autori, između ostalog, obrađuju i teme kao što je promocija fizičke aktivnosti i studentska percepcija igre Pokemon Go, učinci šestonedjeljnih treninga košarkaša na pijesku naspram treninga na parketu, doping i stavovi turskih sportista, uticaj vježbi u bazenu kod starijih žena. Ovo su samo neke od istraživanja na aktuelne teme u oblasti sportskih nauka i medicine, uključujući sve aspekte zdravlja i sporta koje su uključene u ovaj časopis.

Glavni urednici ovog časopisa prof. dr Duško Bjelica i prof. dr Stevo Popović navode da će, prema dosadašnjoj statistici i citiranosti na svjetskom nivou, časopis imati svoj najveći domet u 2020. godini. „MJSSM" časopis će biti ponovo procijenjen od Web of Science, a dosadašnje indikacije ukazuju da će časopis biti uključen u baze ESCI (Science Citation Index Expended) i SSCI (Social Science Citation Index), čime će imati veći nivo uticaja (Bjelica & Popović, 2020).

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Radovi i prezentirani sažeci koji su prijavljeni u okviru konferencije koju organizuje Crnogorska sportska akademija u saradnji sa Fakultetom za sport i fizičko vaspitanje Univerziteta Crne Gore i drugim međunarodnim partnerima, biće objavljeni u supplementu ovog časopisa, nakon sljedeće konferencije. Svrha postojanja ovog časopisa je između ostalog i da se minimiziraju problemi kao što su odlaganje u publikacionom procesu objavljivanja članaka ili prevazilaženja ranijih problema. Stoga, časopis pruža: online otvoren pristup, brzo objavljivanje, mišljenje stručnjaka i istraživača, post publikacijske alate za pokazivanje kvaliteta i učinka, zajednicu zasnovanu na zajedničkom jeziku članaka i svjetsku medijsku pokrivenost.

Metod rada

U ovom radu biće korišćena metoda analize sadržaja, koja se efikasno primjenjuje u raznovrsnim područjima ljudskog rada i stvaralaštva. Ona podrazumijeva korišćenje pisanih izvora kao osnovne jedinice izvora informacija. Jedinica analize sadržaja su naslovi naučnih radova objavljenih u časopisu "Montenegrin Journal of Sports Science and Medicine". Ovdje su to svi naučni članci objavljeni kroz tri broja za 2019. i 2020. godinu časopisa Montenegrin Journal of Sports Science and Medicine.

SLIKA 1. Montenegrin Journal of Sports Science and Medicine, Vol. 9 No. 1 (2020)

U pomenutom časopisu u zadnja tri broja je objavljeno 30 radova, koji će biti klasifikovani prema sljedećim naučnim oblastima: sportska medicina, sportski trening, antropologija spota; psihologija sporta; fiziologija sporta i sistemska analiza radova.

Rezultati

Nakon sprovedene kompletne analize sadržaja iz časopisa Montenegrin Journal of Sports Science and Medicine, izvršena je klasifikacija radova iz šest oblasti, kako je prikazano u Tabeli 1.

Tabela 1. Klasifikacija objavljenih radova prema naučnim oblastima

Naučne discipline	Broj objavljenih radova
Sportska medicina	4
Sportski trening	10
Antropologija sporta	2
Psihologija sporta	7
Fiziologija sporta	4
Sistematska analiza	3
Total	30

Sportska medicina je naučna disciplina koja se bavi primjenom medicine i nauke u sprječavanju, prepoznavanju, liječenju i rehabilitaciji povreda izazvanih sportom, vježbanjem ili rekreativnim aktivnostima. Ova naučna oblast u "Montenegrin Journal of Sports Science and Medicine" časopisu je bila zastupljena kroz sljedeće radove: "Respiratory parameters in elite finn-class sailors". (Pezelj, Milavic, & Erceg, 2019), "Comparison of the static and dynamic balance between normal-hearing and hearing-impaired wrestlers" (Coskun, Unlu, Golshaei, Kocak, & Kirazc, 2019), "Doping Knowledge and Attitudes of Turkish Athletes: A Cross-Sectional Study" (Ozkan et al., 2020), "Manner of Execution and Efficacy of Reception in Men's Beach Volleyball" (Palao, Lopez-Martinez, Valades, & Hernandez, 2019).

Sportski trening kao sastavni dio trenažne tehnologije, omogućavaju transformacione procese pojedinih segmenata antropološkog statusa sportiste i usavršavanje tehničko-taktičkih elemenata, što u krajnjem vodi ka postizanju vrhunskih sportskih dostignuća. Iz ovog veoma važnog segmenta su prikazani sljedeći radovi: "Exponential versus linear tapering in junior elite soccer players: effects on physical match performance according to playing positions" (Krespi, Sporis, & Popovic, 2019), Neuromuscular adaptations after blood flow restriction training combined with nutritional supplementation: A preliminary study" (Chulvi-Medrano et al., 2019), "The Effects of High-Intensity Interval Training on Skeletal Muscle Morphological Changes and Denervation Gene Expression of Aged Rats" (Tayebi, Siahkouhian, Keshavarz, & Yousefi, 2019), "The Association between Relative Age Effect, Goals Scored, Shooting Effectiveness and the Player's Position, and her Team's Final Classification in International Level Women's Youth Handball" (Saavedra, & Saavedra, 2020), "The Effects of A 6-Week Plyometric Training Programme on Sand Versus Wooden Parquet Surfaces on the Physical Performance Parameters of Well-Trained Young Basketball Players" (Ozen, Atar, & Koc, 2020), "The Effect of an In-Season 8-Week Plyometric Training Programme Followed By a Detraining Period on Explosive Skills in Competitive Junior Soccer Players" (Branquinho et al., 2020), "Warm-Up Striding Under Load Does Not Improve 5-Km Time Trial Performance in Collegiate Cross-Country Runners" (O'Neal et al., 2020), "The influence of coaches' instruction on technical actions, tactical behaviour, and external workload in football small-sided games" (Batista et al., 2019), "The important game-related statistics for qualifying next rounds in Euroleague" (Dogan, & Ersoz, 2019) i "Normative Profile of the Efficacy and Way of Execution for the Block in Women's Volleyball from Under-14 to Elite Levels" (Echeverría, Ortega, & Palao, 2020).

Antropologija sporta je naučna disciplina koja se bavi konstrukcijom antropoloških modela, utvrđivanjem strukture i relacija antropoloških karakteristika, kao i utvrđivanjem relacija u odnosu na odgovarajuće kineziološke fenomene. Zastupljeni radovi iz ove oblasti su sljedeći: "Regional differences in adult body height in Kosovo" (Masanovic, Bavcevic, & Prskalo, 2019) i "Investigation of Pool Workouts on Weight, Body Composition, Resting Energy Expenditure, and Quality of Life among Sedentary Obese Older Women" (Rezaeipour, 2020).

Psihologija sporta je nauka koja, objašnjavaja brojna stanja psihe spotrista u različitim okolnostima, položaj jedinke u sportskom kolektivu, odnose pojedinca prema grupi ili šire društvenoj zajednici, kao i zakonitosti unutar složenih psiholoških okvira, zastupljeni su sljedeći radovi: "Fathers – an untapped resource for increasing physical activity among African American girls" (Blackshear, 2019), "The effects of weekly recreational soccer intervention on the physical fitness level of sedentary young men" (Aslan, Salci, & Guvenc, 2019), "Psychological State and Behavioural Profiles of Freshman Enrolled in College and University Instructional Physical Activity Programmes under Different Policy Conditions" (Kim, & Cardinal, 2019), "Factors Affecting Critical Features of Fundamental Movement Skills in Young Children" (Marinsek, Blazevic, & Liposek, 2019), "How Grit is Related to Objectively Measured Moderate-to-Vigorous Physical Activity in School Student" (Hein, Kalajas-Tilga, Koka, Raudsepp, & Tilga, 2019), "Does it Promote Physical Activity? College Students' Perceptions of Pokémon Go" (Yan, Finn, & Breton, 2020) i "Evaluation of Risks and Benefits of Physical Activity of Hypertensives and Normotensives: Fighting a Societal Burden" (Bras et al., 2020).

Fiziologija sporta je integralni dio sportske medicine i bavi se izučavanjem funkcionisanja organskih sistema i organizma u cjelosti prilikom raznovrsnih fizičkih aktivnosti čovjeka. Iz ove oblasti su zastupljeni sljedeći radovi: "Connection in the fresh air: A study on the benefits of participation in an electronic tracking outdoor gym exercise programme" (Johnson, Ivarsson, Parker, Andersen, & Svetoft, 2019), "Examination of Exercise-Induced Skeletal and Cardiac Muscle Damage in Terms of Smoking" (Ipekoglu, Taskın, & Senel, 2019), "Burnout and Coping Strategies among Private Fitness Centre Employees" (Georgiou, & Fotiou, 2019) i "The Effect of 16-Minute Thermal Stress and 2-Minute Cold Water Immersion on the Physiological Parameters of Young Sedentary Men" (Podstawski, Borysławski, Clark, Laukkanen, & Gronek, 2020).

Sistemska analiza radova predstavljaju veoma bitne karike u složenom sistemu društvenih odnosa u sportu. Zastupljena su tri rada u kojima je prikazana zanimljiva hronološka retrospektiva tema koje su se odnosile na pojedina sportska zbivanja, objašnjen je i način sistematizacije analize radova. A radi se o sljedećim radovima: "Research Quality Evaluation in Social Sciences: The Case of Criteria on the Conditions and Requirements for Academic Promotion in Serbia, Slovenia and Montenegro" (Popovic, Pekovic, & Matic, 2019), "The Development of an Online Surveillance of Digital Media Use in Early Childhood Questionnaire- SMALLQ™-For Singapore" (Chia, Tay, & Chua, 2019) i Match Analysis in Handball: A Systematic Review" (Ferrari, Sarmento, & Vaz, 2019).

Diskusija

U ovom radu su klasifikovane teme iz časopisa „Montenegrin Journal of Sports Science and Medicine" koji su objavljeni kroz tri broja za 2019. i 2020. godinu, te na osnovu medote analize sadržaja došli do podatka da su najbrojniji radovi iz oblasti Sportskog treninga (10), zatim Psihologije sporta (7), Sportske medicine (4), Fiziologije sporta (4), Sistematske analize (3) i Antropologije sporta (2).

Kao što je već navedeno, najveći broj istraživanja je iz Sportskog treninga, za koje se može reći da su najviše citirani i najbolje kotirani, što ne čudi s obzirom na veliki opseg ove naučne discipline i sami značaj problematike koju tretira. Sportski trening, kao sastavni dio trenažne tehnologije, omogućava transformacione procese pojedinih segmenata antropološkog statusa sportiste i usavršavanje tehničko-taktičkih elemenata, što u krajnjem vodi ka postizanju vrhunskih sportskih dostignuća (Bjelica & Fratric, 2011; Bjelica, 2013; Bajramovic, Likic, Manic, & Mekic, 2015; Bajramovic et al., 2018). Da je trenažni program rada kod svih naučnih originalnih radova iz oblasti sportskog treninga doveo do pozitivnih transformacija, potvrđuje Vukotic (2018) u svom naučnom radu gdje je izvršila analizu sadržaja objavljenih originalnih naučnih radova iz oblasti sportskog treninga. Takođe, utvrđeno je da su objavljeni radovi u ovom časopisu imali za teme najsavremenije tendencije u sportskim naukama, pa je i to razlog napredovanja časopisa u smjeru najviših naučnih baza u sportskim naukama. Ne može se izostaviti činjenica da se MJSSM časopis i dalje suočava sa velikim uspjehom, i ako je ušao u dvije najprestižnije baze podataka indeksa (Web of Science i Scopus), jedna od tih baza podataka (Scopus) i dalje nastavlja da prepoznaje razvoj ovog časopisa, dok se sa druge strane očekuje da MJSSM uskoro dobije faktor uticajnosti (IF) od strane Web of Science-a i tako uđe na teško dostupnu i prestižnu SCI listu, čime bi zabilježio najveći uspjeh izdavačke djelatnosti u istoriji naučnih časopisa u Crnoj Gori.

Časopis "Montenegrin Journal of Sports Science and Medicine" vezan je za naučnu konferenciju koju Univerzitet Crne Gore i Crnogorska sportska akademija organizuju svake godine početkom aprila, gdje budu ugošćeni i uvaženi respektabilni profesori i naučnici koji rade na volonterskoj osnovi, na recenzijama i uređivanju časopisa. Zapravo, naučni radovi sa konferencije se objavljuju u časopisima "Montenegrin Journal of Sports Science and Medicine", "Sport Mont" i "Journal of Anthropology of Sport and Physical Education", koji su indeksirani u prestižnim svjetskim međunarodnim bazama.

Časopis "Montenegrin Journal of Sports Science and Medicine" vezan je za naučnu konferenciju koju Univerzitet Crne Gore i Crnogorska sportska akademija organizuju svake godine početkom aprila, a koju je UNESKO-ov Internacionalni savjet za sportsku nauku i fizičko vaspitanje proglasio za jedan od najznačajnijih događaja u oblasti sporta za 2019. godinu, koja je okupila je 223 delegata iz gotovo 50 država, dok naučne radove potpisuje 364 autora i koautora (Bavcevic, 2019).

Ono što je važno istaknuti još za časopis jeste da je od 2017. godine urednički odbor časopisa pokrenuo dodjeljivanje nagrade za izuzetne sportske rezultate na međunarodnom planu i promociju crnogorskog sporta. To priznanje do sada je pripalo košarkašu Bojanu Dubljeviću, rukometašici Ani Milačić i karatisti Mariu Hodžiću. Time je "MJSSM" dodatno promovisao svoju ulogu u sportskoj i široj javnosti na unapređenju svijesti o značaju sporta.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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JASPE Journal of Anthropology of Sport and Physical Education

REVIEW PAPER

The Analysis of Published Papers written by the Lecturer of the Faculty of Sport and Physical Education in Sport Mont from 2010 to 2012

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Abstract

Sport Mont is an electronic and printed scientific journal that has been created in 2003 and it has been developed by Montenegrin Sport Academy. Main purpose of this journal is to represent people, the new scientific researches from sport science and medicine. This journal has been publishing 3 times per year, in February, June and October from the 2003 until nowadays. It has been published more than 1400 papers. Right now, Sport Mont is indexed in 23 scientific bases, in which the most popular are Scopus and Web of Science. In this paper, it has been made a systematic review of papers published by the lectures from Faculty from sport and physical education from Niksic in a period from 2010 to 2012. It has been founded 60 papers that were researching an amount of different areas of sport science. The results have showed that the lectures from the Faculty of Sport and Physical Education had the most interest in the area of sport training and from that area it has been founded 28 papers. When we talk about lectures that were publishing in that period, professor Dusko Bjelica has published 18 papers meaning that he has been the one who has published the most. At the end we can tell that the lectures from this faculty became very beneficent for this journal, which it shows their professionality and effort. Also, by putting this journal on the top of the area of sport science and promoting science in general in Montenegro.

Keywords: *Sport Mont, Analysis, Science, Montenegro*

Uvod

„Crnogorska sportska akademija" i njen predsjednik dr Duško Bjelica su pokrenuli 2003. godine Sport Mont: časopis za fizičku kulturu i zdravlje, u kojem sarađuju eminentni sportski naučnici iz zemlje, regiona i Evrope (Slika 1). Časopis „Sport Mont", izlazi tri puta u toku godine, u izdanju Crnogorske sportske akademije. U njemu se mogu pronaći radovi iz oblasti sportskih nauka, medicine i fizičkog vaspitanja.

Od 2003. do danas je objavljeno preko 1000 naučnih radova, od strane stručnjaka iz navedenih oblasti sa svih strana svijeta. Većina ovih radova je prezentovana na naučnim konferencijama koje svake godine tradicionalno održava Crnogorska sportska akademija u saradnji sa svojim partnerima. Kroz svoj dugi put od nastajanja do danas, časopis je prošao kroz brojne transformacije. Danas je indeksiran u 23 međunarodne baze podataka. Glavni urednici časopisa su ugledni profesori i naučnici dr Duško Bjelica sa Fakulteta za sport i fizičko vaspitanje iz Nikšića i dr Zoran Milošević sa Fakulteta za sport i fizičko vaspitanje iz Novog Sada.

U ovom radu su analizirani radovi objavljeni u periodu od 2010. do 2012. godine, a u kojima su autori ili koautori profesori sa Fakulteta za sport i fizičko vaspitanje iz Nikšića. U tom periodu od strane predavača iz Nikšića objavljeno je 60 radova. Radovi su se uglavnom bavili problematikom sportskog treninga i antropomotorike. Čak polovina radova pripada upravo toj oblasti i ima ih 28. Takođe, mogu se pronaći radovi i iz biomehanike, menadžmenta, ali i ostalih oblasti. Tako se već može steći utisak o jednom širokom polju interesovanja od strane predavača iz Nikšića, koji svojim velikim zalaganjem i trudom, unapređuju časopis,

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kako bi ga doveli do samog vrha kada je u pitanju sportska nauka. Radovi ovog tipa su velikog značaja, jer prije svega na jednom mjestu prikupljaju i kategorizuju radove, što nekim slijedećim istraživačima čini veliku uslugu kada je u pitanju pronalaženje radova za neka nova istraživanja. Isto tako predstavljaju i jednu vrstu rekapitulacije onoga što se istraživalo u prethodnom periodu i kao jedan mali podsjetnik, konkretno ovaj rad daje lijepu sliku o jednom velikom trudu i zalaganju. To svakako daje i jednu vrstu motivacije da se na tome ne stane, već da se ide naprijed i stvaraju neke nove granice u svijetu nauke.

SLIKA 1. Časopis Sport Mont Vol.10 No. 34-35-36 (2012)

Metod

Za ovaj tip istraživanja biće korišćena metoda analize sadržaja. Jedinica koja se analizirala u ovom radu su bili naslovi radova objavljenih od strane profesora sa Fakulteta za sport i fizičko vaspitanje iz Nikšića, koji su učestvovali kao autori ili koautori u istim, u periodu od 2010. do 2012. godine. Radovi su pažljivo pregledani i klasifikovani po grupama, naučnim disciplinama u okviru sportskih nauka. Iz perioda od 2010. do 2012. pronađeno je 60 radova u kojima su profesori sa Fakulteta za sport iz Nikšića učestvovali kao autori ili koautori. Radovi su klasifikovani po disciplinama u okviru sportske nauke i to na: sportski trening i antropomotorika, biomehanika, sportska medicina i fiziologija, sociologija, psihologija i menadžment u sportu, teorija sporta, istorija sporta, antropologija, metodologija istraživanja. Detaljnija analiza je prikazana u narednom dijelu teksta.

Rezultati

Analizom radova, došlo se do zaključka da su se u tome periodu profesori iz Nikšića najviše bavili problematikom sportskog treninga i antropomotorike. S toga je najviše radova upravo iz tih oblasti. To su slijedeći radovi: "The influence of strength training of growth and development of subadolescent and early adolescent children" (Radulovic & Gojkovic, 2010), "Influence of summer break at some motoric abilities on football players aged 10 years" (Popovic, Molnar, & Smajic, 2010), "Comparation some motoric abilities two generation of football school players" (Molnar, Popovic, & Doder, 2010), "Dijagnostika nivoa treniranosti posebnih populacija na bazi parametara frekvence srca i brzine trčanja na anaerobnom pragu" (Goranović i Gardašević, 2010), "Research methods in sport training" (Bjelica, 2010), "New standards in modelling top sprinters" (Idrizovic, 2010), "The influence of the sports training on the development of some specific physical abilities of the female handball players" (Vujovic, & Petkovic, 2010), "Differences of motorical abilities in relation to sex of pupils" (Gojkovic, 2010), "Motoric status relations in montenegrin yuoth population of various sport orientations" (Bjelica & Petkovic, 2010), "Analysis of differences between boys attending a football school and thos who do not do sport with specific-motor abilities" (Molnar, Smajic, Popovic, & Tomic, 2010), "Motor abilities of female students with regard to their work conditions" (Mitrevski, Georgiev, & Hadzic, 2011), "Dijagnostika anaerobnog praga u funkciji ocjene aerobne izdržljivosti posebnih populacija" (Goranović, 2011), "Situacioni trening u fudbalu" (Aćimović, Hadžić i Spirtović, 2011), "Streinght in swimming" (Madic, Okicic, Rasovic, &, Okicic, 2011), "The effects of programmed work in preparation period of kadet football players on their explosive force transformation" (Gardasevic & Goranovic, 2011), "Differences of anthropometric characteristic and motor abilities of different sport orientation" (Vukotic, 2011), "Differences of motor and functional abilities in football and handball players aged thirteen to fifteen years" (Vukotic & Musovic, 2011), "Development of streinght in annual cycle training" (Rasovic, Madic, Okicic, Okicic, & Petrovic, 2011), "Resulting efficacy of mid-track race in modern olympism" (Goranovic & Gardasevic, 2011), "Influence of motor manifestations from the eurofit program for children on motor skills and habits at high school female students" (Mitrevski, Georgiev, & Hadzic, 2011), "Diagnostics of motor ability as a base of correction planning of transformation processes in special populations" (Goranovic, 2011), "The methodology of shaping situational training in the case of alpine skiing" (Joksimovic, Acimovic, & Hadzic, 2011), "Struktura parcijalizovanog motoričkog prostora učenika adolescentne dobi" (Idrizović, 2011), "Influence of motor capabilities on efficacy in performing wedge turning in alpine skiing" (Hadzic, Vujovic, & Muratovic, 2012), "The influence of rhytmic gymnastics teaching contents upon development of some motor skills among the schoolgirls of the fifth grade of primary school" (Vujovic, 2012), "Validity of the situational-motor tests with football players at the age of 15" (Gardasevic & Bjelica, 2012), "Basic motor abilities and situational motor efficiency with young handball players from Montenegro - quantitative approach" (Muratovic & Georgiev, 2012), "Thermovision application in kendo training" (Roglic, Fratric, Nesic, Bjelica, & Madic, 2012).

Iz oblasti biomehanike su slijedeći radovi: "Rationalization of bipedal movement – speed –" (Bjelica, 2010), "Elementary techniques of basic hand strokes in modern karate" (Radovanovic & Popovic, 2011) "Frequency and structure of sagittal disorders of the spinal column wieh adolescent students" (Jovovic, 2010), "Flagellum" efekat u sportu" (Opavsky, 2012).

Iz oblasti medicine i fiziologije sporta su pronađeni sljedeći radovi: "The influence of autonomous diving on senses and mental processes" (Krivokapic, 2010), "Povratak sportskim aktivnostima poslije rupture Ahilove tetive" (Kezunović, 2010), "Kompresija supraskapularnog nerva kod odbojkaša" (Kezunović i Laković, 2010), "Kampanja o antidopingu u crnogorskom sportu i njeni rezultati" (Stijepović, Ćeranić i Kezunović, 2010), "Coliosis like the deformity of the spinal cord and the ability for sports activies" (Joksimovic, Joksimovic, & Vujovic, 2010), "Ankle sprain: who is most frequently injured and how long athletes are absent from the field?" (Cvejanov-Kezunovic, Kezunovic, Popovic, & Bjelica, 2011), "The effects of natural stimulants on the human organism" (Petkovic & Krivokapic, 2012).

Radi lakše kategorizacije, slijedeća grupa radova je iz sociologije, psihologije i sportskog menadžmenta: "Development and shaping of students' intrinsic motivation in physical education" (Krivokapic & Krivokapic, 2010), "Improvement of health and life quality in population through social support for development of phycical culture" (Krivokapic & Krivokapic, 2010), "Obilježja sportske propagande" (Bjelica, 2010), "The attitudes toward sport advertising among the question how often consumers purchase sporting goods" (Popovic, 2011), "The attitudes toward sport advertising among the question how often consumers participate in sports activities" (Popovic, Molnar, & Radovanovic, 2011), "The olympic games as actuator of an international cooperation and strengthening of interstate friendly relations" (Simonovic & Krivokapic, 2012), "Usage of athletes as endorsers" (Popovic, 2012), "Qualification structure of human resources in sport in municipalities Bijelo Polje, Mojkovac i Kolasin" (Jovanovic, 2012).

Radovi iz oblasti fizičkog vaspitanja, teorije sporta i istorije sporta su: "The model of the estimation of the erudition-social characteristics of the physical education" (Milosevic, 2010), "Montenegro and olympism" (Rasovic, 2010), "Antičke olimpijske igre i moderni olimpizam" (Goranović i Bjelica, 2010), "Target planning of the physical educationt" (Milosevic & Bulatovic, 2010), "Physical activity in medical vocational and training school students and their attitude on its significance" (Joksimovic, Joksimovic, & Vujovic, 2010), "Supplement to humanistic concept of youths' sports" (Bjelica & Krivokapic, 2011).

Iz antropologije su objavljena dva rada: "The differences in body composition between football players of different rank competitions" (Popovic, Smajic, Joksimovic, & Masanovic, 2010), "Correlation of the morphological characteristics and sports achievements in karate" (Bjelica & Petkovic, 2012).

Metodologija istraživanja ima jedan rad: "Biodinamička metodologija u sportu" (Opavsky i Bjelica, 2011).

Ostali radovi koji su objavljeni u periodu od 2010. do 2012. a nisu kategorizovani su: "Dalji pravci razvoja crnogorske sportske akademije" (Nikolić i Bjelica, 2011), "Forgotten mission of physical education" (Milosevic, Maksimovic, Matic, & Bjelica, 2011), "Effective knowledge tests and feedback in function of improving the students' study" (Bjelica & Krivokapic, 2011), "Attitudes of montenegrin professors of physical education in the field of personal specialization, work satisfaction and sports development in Montenegro" (Bjelica & Krivokapic, 2012). Razlog zbog kojeg nisu kategorizovani je taj da prethodna četiri rada sadrže više oblasti, tj. mogu se naći u više disciplina u okviru sportskih nauka. U Tabeli br.1 prikazane su naučne discipline i broj radova u okviru njih.

Tabela 1. Broj radova po naučnim disciplinama

Naučna disciplina	Broj objavljenih radova
Sportski trening i Antropomotorika	28
Biomehanika	4
Sportska medicina i fiziologija	7
Sociologija, psihologija, menadžment u sportu	8
Teorija sporta, Istorija sporta	6
Antropologija	2
Metodologija istraživanja	1
Ostali radovi	4

Diskusija

U ovome radu analizirana je aktivnost predavača sa Fakulteta za sport i fizičko vaspitanje koji su objavili radove u časopisu „Sport Mont" u periodu od 2010. do 2012. godine. U radovima koji su navedeni, predavači sa Fakulteta za sport iz Nikšića su učestvovali kao samostalni autori, autori ili koautori. U prethodnom dijelu teksta, radovi su klasifikovani po naučnim oblastima koje su u okviru sportskih nauka.

Kao što može da se primjeti, za taj period najviše radova je objavljeno iz oblasti sportskog treninga i antropomotorike kojih ima 28, dakle, skoro polovina radova koji su objavljeni u tom periodu od strane predavača iz Nikšića. Iz biomehanike su objavljena 4 rada. Sportska medicina i fiziologija ima 7 objavljenih radova. Isto tako, oblasti koje imaju veliki broj radova su one koje su se bavile, sociologijom i psihologijom sporta, kao i sportskim menadžmentom koji se može tumačiti i kao jedna vrsta sociologije sporta. Radova iz ovih oblasti ima 10. Šest radova je objavljeno iz oblasti teorije fizičkog vaspitanja i sporta i istorije sporta, od čega su 4 rada iz teorije sporta i fizičkog vaspitanja i 2 rada iz istorije sporta. Dva rada su objavljena iz antropologije kao i jedan rad iz oblasti metodologije istraživanja.

Kada su u pitanju profesori koji su u tome periodu objavljivali radove, profesor Duško Bjelica ima najviše objavljenih radova (17), od toga 3 rada kao samostalni autor, 4 rada kao autor i 8 radova u kojima je učestvovao kao koautor. Nakon njega slijedi profesor Dragan Krivokapić sa 10 objavljenih radova (1 kao samostalni, 9 kao koautor), a odmah iza Krivokapića slijedi profesor Stevo Popović sa 9 radova (2 kao samostalni, 3 kao autor, 4 kao koautor). Ostali profesori su imali po 6 ili manje radova u kojima su učestvovali kao samostalni autori, autori ili kao koautori. Kao što se može zaključiti, veliki je broj radova objavljenih od strane predavača sa Fakulteta za sport iz Nikšića. Takođe, ne samo da je njihova aktivnost bila velika već te radove krasi i jedna raznovrstnost, jer su se bavili raznim disciplinama. Nakon svega navedenog, zaključuje se da je angažovanost predavača sa FSFV iz Nikšića izuzetno velika i profesionalna. Naravno, sve to doprinosi razvoju sportskih nauka u Crnoj Gori. Kada se još uzme u obzir da se časopis „Sport Mont" nalazi u svjetskim naučnim bazama, to naravno daje još više na značaju i pokazuje maksimalnu profesionalnost predavača sa Fakulteta za sport i fizičko vaspitanje iz Nikšića.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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SHORT REPORTS

Association between Physical Activity Level and Hemoglobin Concentration in Male College Students

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Abstract

Increased physical activity is useful for reducing the risk of most non communicable diseases. However, the effect of high physical activity on hematological profile is not established yet. The aim of this study is to investigate the association of physical activity and hemoglobin concentration in college students. This cross-sectional study involved 67 male students of Faculty of Medicine and Health Science, Atma Jaya Catholic University of Indonesia. The physical activity was evaluated using International Physical Activity Questionnaire (IPAQ). Hemoglobin concentration was measured using a hemoglobinometer. Anova test with LSD post hoc analysis was applied to compare the hemoglobin concentration between groups. Significance was set at $p<0.05$. There were 17 students (25.4%) with low physical activity, 20 students (29.8%) with moderate physical activity, and 30 students (44.8%) with high physical activity. The hemoglobin concentrations were 15.6g/dL in low physical activity, 14.8 g/dL in moderate physical activity, and 15.2 g/dL in high physical activity. LSD post hoc analysis showed that hemoglobin in low physical activity is significantly higher than in moderate ($p=0.027$). Students with moderate physical activity had the lowest hemoglobin concentration, and significantly lower than students with low physical activity.

Keywords: *Physical Activity Level, College Students, Hemoglobin Concentration, Anemia*

Introduction

Physical inactivity has become a major concern worldwide. A study involving large amount of participants over 15 years old, reported that about 20% did not engage in sufficient physical activity (Dumith, Hallal, Reis, & Kohl, 2011). A study in US also reported that large portion of population had insufficient amounts of physical activity (Tucker, Welk, & Beyler, 2011). Physical inactivity is also commonly found in young people. Concern is increasing due to physical inactivity is also commonly found in young people. A meta-analysis study revealed that about 40%-50% college students did not have sufficient physical activity (Keating, Guan, Piñero, & Bridges, 2005). In addition, most people spend their average of 8 hours per day taking part in sedentary behaviors (Schuna Jr, Johnson, & Tudor-Locke, 2013).

Physical inactivity raises many problems on health, especially on non-communicable diseases. Sedentary behaviors are attributed to increased risk of metabolic syndrome in young adults (Salonen et al, 2015). A study by Warren et al reported men who drove car 10 hours or more per week had a greater possibility of CVD mortality than those who drove fewer than 4 hours per week (Warren et al, 2010). Physical inactivity is also related with malignancy. Shoemaker et al reported physical inactivity as a modifiable risk factor for cancer in young adult both male and female (Shoemaker et al., 2017). Sedentary behavior is considered as the fourth highest risk factor leading to death after hypertension, tobacco use, and high blood glucose (WHO, 2018).

Otherwise, it has been reported widely the numerous advantages of increased physical activity on health. The benefits of exercise have been documented, including reducing the risk for all-cause mortality, heart disease, high blood pressure, stroke, type 2 diabetes and some cancers (Kushi et al, 2012). The effects of exercise on body's organ system are usually positive and in single form. For example, the effect of exercise on vascular system is to diminish peripheral resistance and to lower blood pressure. There has been no evident an

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inverse effect. However, exercise has a distinct effect on hematologic profile. A study by Pourghardash & Nikseresht reported that aerobic exercise increased hemoglobin (Hb) concentration and hematologic factors in young females (Pourghardash & Nikseresht, 2017). The increased Hb lead to increase oxygen carrying capacity and maximal oxygen capacity. In the other hand, Çiçek (2018) found an opposite result in his study. Exercise decreased red blood cells (RBC), hemoglobin (Hb), and hematocrit (Hct) in sedentary women (Çiçek, 2018). This conflicting result is an interesting subject to study. Thus, the purpose of this study was to find an association between level of physical activity and hemoglobin concentration.

Methods

This cross-sectional study was conducted on 67 students of Faculty of Medicine and Health Science, Atma Jaya Catholic University of Indonesia. Only male students were invited to follow the study. Exclusion criteria consisted of: tobacco use, having chronic blood diseases or other chronic diseases, consumption of blood booster vitamins or supplements as well as drugs affecting hematologic profile, and under blood administration therapy. Each participants gave their informed consent after receiving a written information. The study was approved by the ethics commission of Faculty of Medicine, Atma Jaya Catholic University of Indonesia, Jakarta, Indonesia(No: 22/12/KEP-FKUAJ/2017).

Level of physical activity was assessed by a short version of International Physical Activity Questionnaire (IPAQ). The IPAQ is applicable for people aged 15-69 years old. Subjects were asked to record their kinds of physical activity for last 7 days which is classified into vigorous, moderate, walking, and sitting. Frequency (day/week) and duration (minutes) of each classification is recorded. The intensity of physical activity is expressed in MET-minutes a week. Level of physical activity was categorized into high, moderate and low according to the scoring rule of IPAQ (Cleland, Ferguson, Ellis, & Hunter, 2018).

Hemoglobin was analysed using a portable device (Quik-Check® hemoglobinometer, London). Hemoglobinometer was calibrated prior to the study according to the factory manual handling. The device requires only 10 μL blood from peripheral capillaries. The results is obtained in 15 seconds with wide Hb measurement range of 4.5-25.6 g/dL

The numerical data was presented as mean ± standard deviation (SD) while categorical data as frequency and percentage. Anova test continued with LSD post hoc analysis was applied to evaluate Hb concentration between groups of physical activity level. Statistical analysis was processed by using the statistical package program STATA 14.2 (StataCorp, Texas, USA). P value of <0.05 was considered significant.

Results

Subjects' characteristics are presented in table 1. Most participants had high and moderate physical activity level (44.8% & 29.8%), while only 25.4% had low physical activity.

Table 1. Characteristics of the participants

Variables	Mean±SD
Age (years)	19.9±0.8
Physical activity level (MET min a week)	
Low (n=17)	346.2±205.2
Moderate (n=20)	1170.4±248.3
High (n=30)	3516.3±1271.1
Total mean	2011.6+1641.9
Hemoglobin (g/dL)	15.2±1.1

Note: MET - metabolic equivalent; Mean – arithmetic mean; SD – standard deviation

Table 2 demonstrates the distribution of participants according to the physical intensity. Subjects might be involved in more than one type of activity. There were only small numbers of students participated in vigorous activity 3 days or more (19.4%) and much smaller in moderate activity for 5 days or more in a week (3%). As many as 49.3% students reported walking more than 5 days a week.

Table 2. Physical activity level and type of activity intensity

Intensity	Frequency (%)
Level of physical activity	
Low	17 (25.4%)
Moderate	20 (29.8%)
High	30 (44.8%)
Vigorous	
No	38 (56.7%)
<3 days	16 (23.9%)
3 days or more	13 (19.4%)
Moderate	
No	42 (62.7%)
<5 days	23 (34.3%)
5 days or more	2 (3.0%)
Walking	
<5 days	34 (50.7%)
5 days or more	33 (49.3%)

The hemoglobin concentration between groups of physical activity level was compared (table 3). Anova test indicated that there was significant difference at least between two groups (p=0.04). Post hoc analysis using LSD revealed that Hb level in low physical activity was significantly higher than in moderate physical activity (p=0.027). The Hb level between low physical activity and high physical activity, and between moderate physical activity and high physical activity were not significantly different (p=0.139 and p=0.319, respectively).

Table 3. Comparison of hemoglobin level between groups

Level of physical activity	Hb concentration (g/dL)	95% CI	p
Low physical activity	15.7±1.1	15.1-16.2	
Moderate physical activity	14.8±1.0	14.4-15.3	.040
High physical activity	15.2±1.2	14.7-15.6	

Note: Hb - hemoglobin; CI - confident interval.

Discussion

Study on the effect of exercise on hemoglobin concentration is fascinating due to distinct results between studies. Researchers are challenged to conduct a study on that topic to confirm and to compare the results of previous studies. Our study also attempted to reveal the association between level of physical activity and hemoglobin concentration in male college students. The results showed that students with low physical activity had the highest Hb level and significantly higher compared to those with moderate physical activity.

The influence of exercise on Hb concentration has been hypothesized. Exercise, especially aerobic or endurance training increases red blood cells volume. However, increased Hb does not occur due to higher increased of plasma volume which also happens following exercise (Wilmore, Costill, & Kenney, 2008). Instead, increased plasma volume exceeding increased red blood cells causes a decrease in Hb, resulting in a slight reduces in hematocrit and creates a relatively lower concentration of Hb called pseudoanemia (Wilmore, Costill, & Kenney, 2008).

Many studies have shown a low Hb value due to the effect of exercise. A study by Sharpe et al (2002) reported that hematocrit values below 44% was found in more than four fifth of the female athletes and one fifth of male athletes (Sharpe et al, 2002). Heinicke et al (2001) also reported an inverse correlation between hematocrit and training status in elite athletes, in which the more trained the athletes, the lower the hematocrit. A study by Alam et al (2014) demonstrated that intense exercise could reduce the hemoglobin concentration and red blood cells volume significantly compared to the control group. Regarding the effect of type of exercise on blood profile, an investigation by Çiçek (2018) found hemoglobin, hematocrit, and red blood cells reduced significantly after strength training compared to aerobic training. Results of our study supported those previous studies. Hemoglobin concentration in subjects with moderate physical activity was significantly lower than those with low physical activity.

Some previous studies reported an opposite result. Sazvar et al investigated the effect of aerobic training three times a week for 8-weeks. The results showed that the total number of red blood cells and hemoglobin concentration increased significantly (Sazvar, Mohammadi, Nazem, & Farahpour, 2013). Results from study by Pourghadash et al also established the increased hemoglobin concentration following 8-week of aerobic exercise (Pourghardash & Nikseresht, 2017). A study in women with rheumatoid arthritis (RA) also found hemoglobin, hematocrit and red blood cells slightly increased after 8-week exercise (Shapoorabadi, Vahdatpour, Salesi, & Ramezanian, 2016). Several factors are attributed with different changes of blood profile in response to exercise. In the other hand, higher Hb concentration in high physical activity is not easily understood. Allegedly, increased Hb in those with high physical activity is related to an increase in circulating red blood cells due to splenic contraction (Piccione, Casella, Panzera, Giannetto, & Fazio, 2012).

Physical inactivity has become an important issue worldwide. The devastating consequences of inactivity and sedentary makes everyone aware of action to improve physical activity especially in young people. A complete understanding of exercise is required to minimize adverse effect of exercise. People must know the hematological change following exercise is physiological and no adverse effect on health to worry about.

Some limitations are found in this study. Dietary habit was not recorded as hemoglobin was affected by food nutrients. Other hematologic parameters were not examined as they were required to obtain a comprehensive understanding on the effect of exercise. Sample size might be too small to obtain a more reliable level of physical activity. Also, self-evaluation of physical activity is less valuable than exercise intervention in an experimental study.

In conclusion, physical activity had an association between with hemoglobin concentration in college students. We recommend performing further studies by including dietary habit and other hematologic parameters in a larger sample.

Conflicts of Interest

Authors declare there is no conflict of interest in any form.

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Journal of Anthropology of Sport and Physical Education

SHORT REPORTS

Emergency Preparedness for Sudden Cardiac Death and Exertional Heat Stroke during Road Races in China

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Abstract

This short report discusses two deaths that occurred on October 20th 2019 during half-marathons in People's Republic of China. The two fatal events were reported to be the consequence of sudden cardiac death. In September 2019, the State Council of China called for nationwide mass sports participation, which raised urgent operational challenges for event organizers and the sports medicine community in China. This report examined emergency preparedness experiences of road races from the United States, in the hope that the Chinese medicine community could continue to improve their ability to respond, and ultimately, to uphold safe sports participation in China.

Keywords: *Marathon, Race Medicine, Cardiac Arrest, Cold Water Immersion*

Introduction

Sudden cardiac death (SCD) during road races is unusual phenomenon in the United States. Among 10.9 million registered marathon and half-marathon race participants from January 2000 to May 2010, the overall incidence rate of SCD was 0.39 per 100,000 runners (Kim et al., 2012). This incidence rate was further reduced during half-marathons (0.25 per 100,000 runners) than during marathons (0.63 per 100,000 runners) (Kim et al., 2012).

In contrast, the occurrence of SCD during half-marathons seems materially high in China. On October 20th 2019, four runners simultaneously collapsed within 100 meters from the finish line of the Jing Zhou Half-Marathon (Ma & Yu, 2019). Within one minute of the incidence, paramedics attempted to resuscitate the 50-yr-old male using automated external defibrillators. The runner showed no pulse and was pronounced dead while being transported to the hospital. On the same day, one male runner collapsed unconsciously at nearly 20.5 km of the Long Kou Half-Marathon (Ma & Yu, 2019). The runner never recovered consciousness even after three defibrillations and cardiopulmonary resuscitation. According to the incidence report (Ma & Yu, 2019), this runner showed early signs of cardiac arrest and a volunteer medical staff had suggested he stop running, but the runner continued. Had the runner understood the symptoms and the danger of SCD, or had the race organizer had legal authority to withdraw any runner from the race without legal liability, this runner might have survived.

Methods

In September 2019, China's State Council unveiled a document entitled "Opinions on Promoting Mass Sports, Sports Consumption and High-Quality Development of Sports Industry" (State Council, 2019) for promotion of market-based actions to enrich mass sports events. This however raises urgent operational challenges for event organizers and the sports medicine community in China.

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Results and Discussion

First, any successful mass sports event is not possible without organizers' efforts to implement and sustain emergency preparedness programs. In the United States, emergency managers of long-distance events have been continuously refining protocols for emergency events which has resulted in safer sports competition. For example, in the Twin Cities and Marine Corps marathons, the incidence rate of SCD decreased from 1.8 per 100,000 runners during 1976–1994 to 0.45 per 100,000 runners during 1995–2004 (Roberts & Maron, 2005).

Although China has successfully organized a number of international marathons, both national and regional organizers should keep pace with best practices in the face of the rapid growth of marathon events in China. Recent collaboration between the World Athletics and the International Institute of Race Medicine (World Athletics, 2019) offers medical training sessions worldwide for road race medical directors and other healthcare professionals. National administrators and medical leaders of road races in China could learn the latest medical science and trainings from these leading organizations and improve the emergency preparedness capabilities. Local organizers should assemble medical command teams of cardiovascular specialists, emergency physicians, and sports scientists and work with area police departments and hospitals to ensure that resources are in place to handle all anticipated medical needs.

Second, despite a lack of evidence that pre-participation screening of cardiovascular risk factors improves exercise safety, proper education of marathon participants and organizers about individual medical history and symptoms of cardiovascular disease is still warranted, especially for those 1st time participants who are unaccustomed to extended vigorous exercise. One of the important medical lessons from the sports medicine community in the United States has been that most American people (i.e., 95.5% of women and 93.5% of men) aged ≥40 years would be advised to consult a physician before engaging in any form of exercise (Whitfield, Pettee Gabriel, Rahbar, & Kohl, 2014).

The aforementioned tragic events highlight the importance of risk stratification and education among the general Chinese population who now enjoy increased access to marathon participation. The sports medicine community in China could adopt the knowledge recommended by the American College of Sports Medicine (Riebe, Ehrman, Liguori, & Magal, 2017) when identifying risks of exercise participation. Education could be improved using mandatory pre-race training during registration, printed medical information at race sites, and planning training for volunteers.

Third, it is important to recognize that incidences of life-threatening events in the marathon also include non-cardiac clinical origins, including hyponatremia and hyperthermia. In the Boston Marathon, hyponatremia occurs in a substantial fraction of non-elite runners (i.e., racing time of >4:00 hours) (Almond et al., 2005) and proper practice (Noakes et al., 2005) may avert preventable deaths in susceptible runners (e.g., females) (Almond et al., 2005).

Likewise, exertional heat stroke (EHS) pose a threat to the life of competitors in endurance sports performed in mild to hot environments. Fatal events as a result of EHS could be 10 times higher than those of cardiac origins (Yankelson et al., 2014). In 2019, there have been numerous reports of heat illness injury during long-distance events in China (Sina Sports, 2019). On June 6th 2019, one 27-yr-old male runner resumed running following heat exhaustion, eventually collapsed again just 500 meters from the finished line and died after two days of intensive hospital care (Sohu, 2019). The final diagnosis was EHS progressing to multiorgan failure and muscle necrosis. Policymakers in the United States have recognized the risks of EHS and designed educational materials (Casa et al., 2015), which could also be used by the sports medicine community in China to improve EHS diagnosis and injury prevention. Meanwhile, marathon organizers in China should follow validated clinical procedures (Belval et al., 2018), and most importantly, institute the principle of "cool first (i.e., use cold water immersion to reduce the rectal temperature below 38.6°C as quickly as possible), transport second (i.e., then transport the patient to the emergency department)", to improve survival rate of EHS (Demartini et al., 2015).

Until detailed guidelines for pre-participation screening, along with a very carefully coordinated emergency management protocol and repeated training protocols are implemented, we remain cautious about mass marathon participation in China.

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Conflict of Interest
The authors declare that there are no conflicts of interest.

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1.6. After Acceptance

After the manuscript has been accepted, authors will receive a PDF version of the manuscripts for authorization, as it should look in printed version of JASPE. Authors should carefully check for omissions. Reporting errors after this point will not be possible and the Editorial Board will not be eligible for them.

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1.7. Code of Conduct Ethics Committee of Publications

C O P E COMMITTEE ON PUBLICATION ETHICS

JASPE is hosting the Code of Conduct Ethics Committee of Publications of the **COPE** (the Committee on Publication Ethics), which provides a forum for publishers and Editors of scientific journals to discuss issues relating to the integrity of the work submitted to or published in their journals.

2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. *See* example:

Analysis of Dietary Intake and Body Composition of Female Athletes over a Competitive Season

Original Scientific Paper

Diet and Body Composition of Female Athletes

Svetlana Nepocatych[1], Gytis Balilionis[1], Eric K. O'Neal[2]

[1]Elon University, Department of Exercise Science1, Elon, NC 27215
[2]University of North Alabama, Department of Health, Physical Education and Recreation, Florence, AL 35632

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2525 CB

Elon, NC 27244

United States

E-mail: snepocatych@elon.edu

Word count: 2,946

Word count: 4259

Abstract word count: 211

Number of Tables: 3

2.1.1. Title

Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: "and", "or", "between" etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

2.1.2. Type of publication

Authors should suggest the type of their submission.

2.1.3. Running head

Short running title should not exceed 50 characters including spaces.

2.1.4. Authors

The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

In case all authors belong to the same institution affiliation numbering is not needed.

Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

2.1.6. Corresponding author

Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

2.1.7. Manuscript information

All authors are required to provide word count (excluding title page, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References), the Abstract word count, the number of Tables, and the number of Figures.

2.2. Abstract

The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

Since abstract is independent part of your paper, all abbreviations used in the abstract should also be explained in it. If an abbreviation is used, the term should always be first written in full with the abbreviation in parentheses immediately after it. Abstract should not have any special headings (e.g., Aim, Results…).

Authors should provide up to six key words that capture the main topics of the article. Terms from the Medical Subject Headings (MeSH) list of Index Medicus are recommended to be used.

Key words should be placed on the second page of the manuscript right below the abstract, written in italic. Separate each key word by a comma (and a space). Do not put a full stop after the last key word. *See* example:

Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

2.3. Main Chapters

Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements (optional), Conflict of Interest (optional), and Title, Author's Affiliations, Abstract and Key words must be in English (for both each chosen language of full paper). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

2.3.1. Headings

Main chapter headings: written in bold and in Title Case. *See* example:

- ✓ **Methods**

Sub-headings: written in italic and in normal sentence case. Do not put a full stop or any other sign at the end of the title. Do not create more than one level of sub-heading. *See* example:

- ✓ *Table position of the research football team*

2.3.2 Ethics

When reporting experiments on human subjects, there must be a declaration of Ethics compliance. Inclusion of a statement such as follow in Methods section will be understood by the Editor as authors' affirmation of compliance: "This study was approved in advance by [name of committee and/or its institutional sponsor]. Each participant voluntarily provided written informed consent before participating." Authors that fail to submit an Ethics statement will be asked to resubmit the manuscripts, which may delay publication.

2.3.3 Statistics reporting

JASPE encourages authors to report precise p-values. When possible, quantify findings and present them with appropriate indicators of measurement error or uncertainty (such as confidence intervals). Use normal text (i.e., non-capitalized, non-italic) for statistical term "p".

2.3.4. 'Acknowledgements' and 'Conflict of Interest' (optional)

All contributors who do not meet the criteria for authorship should be listed in the 'Acknowledgements' section. If applicable, in 'Conflict of Interest' section, authors must clearly disclose any grants, financial or material supports, or any sort of technical assistances from an institution, organization, group or an individual that might be perceived as leading to a conflict of interest.

2.4. References

References should be placed on a new page after the standard title written in upper and lower case letters, bold.

All information needed for each type of must be present as specified in guidelines. Authors are solely responsible for accuracy of each reference. Use authoritative source for information such as Web of Science, Medline, or PubMed to check the validity of citations.

2.4.1. References style

JASPE adheres to the American Psychological Association 6th Edition reference style. Check "American Psychological Association. (2009). Concise rules of APA style. American Psychological Association." to ensure the manuscripts conform to this reference style. Authors using EndNote® to organize the references must convert the citations and bibliography to plain text before submission.

2.4.2. Examples for Reference citations

One work by one author
- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

Works by two authors
- ✓ Duffield and Marino (2007) studied
- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.
- ✓ First citation: Bangsbo, Iaia, and Krustrup (2008) stated that
- ✓ Subséquent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year
- ✓ Krustrup et al. (2003) studied
- ✓ In one study (Krustrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)
- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

Nepocatych, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine, 6*(2), 57-65. doi: 10.26773/mjssm.2017.09.008

Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology, 100*(6), 727-735. doi: 10.1007/s00421-007-0468-x

Krustrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise, 35*(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies, 50*(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia, 4*. Retrieved November 15, 2005, from http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html

Conference paper:

Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf

Encyclopedia entry (print, with author):

Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from http://www.bartleby.com/65/et/ethnolog.html

Thesis and dissertation:

Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport.* Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See* example:

- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. $^{a, b, c}$), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript ^{a}.

- ✓ ^{a}One participant was diagnosed with heat illness and n = 19.^{b}n =20.

Probability notes provide the reader with the results of the texts for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

- ✓ *P<0.05,†p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See* example:

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓ ....in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. JASPE prefers TIFF, EPS and PNG formats.

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Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of JASPE.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written bellow the figure, in sentence case. *See* example:

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, *p>0.01, †p>0.05.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See* example:

- ✓ Figure 1
    - × figure 1
    - × Figure 1.
    - ✓ ….exhibit greater variance than the year before (Figure 2). Therefore…
    - ✓ ….as shown in Figures 1 and 3. (citing more figures at once)
    - ✓ ….result has shown (Figures 1-3) that... (citing more figures at once)
    - ✓ ….in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See* example:

- ✓ Figure 1a
- ✓ …in Figures 1a and b we can…
- ✓ …data represent (Figures 1a-d)…

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10º	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 º	× 10kg	× 12 : 2	× .056

Signs should be placed immediately preceding the relevant number.		
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age
× 45 ± 3.4	× p < 0.01	× males > 30 years of age

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. *See* example:

- ✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*

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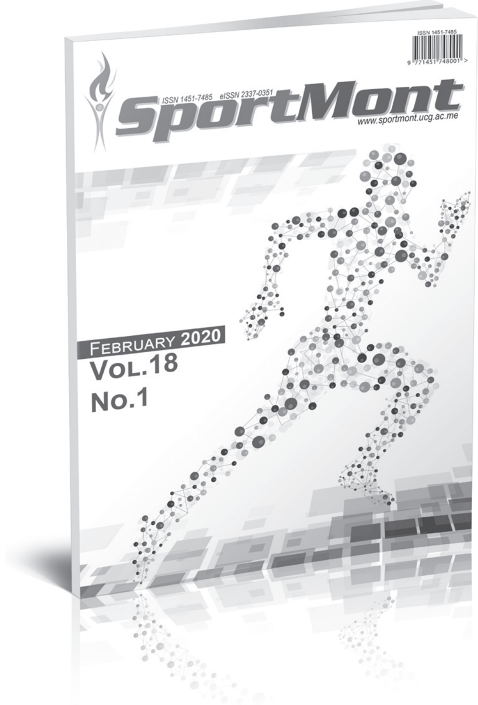

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The University of Montenegro has unique academic, business and development objectives. It comprises 19 faculties and two research institutes. The seat of the UoM is in Podgorica, the capital city, while university units are located in eight Montenegrin towns. The University support services and centers (advisory services, accounting department, international cooperation, career orientation) are located in the Rectorate.

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In 2015/16 there were a total of 1.192 employees at UoM, 845 of which were engaged in teaching. In the same year there were 20.236 students registered at all three cycles of studies.

Internationalization is high on the agenda of UoM priorities, thus it has participated in a number of international projects – over 50 projects funded under the Tempus programme, over 15 Erasmus Mundus Action 2 projects for student mobility, a number of projects under FP7 funding scheme or IPA supported projects, Erasmus + capacity building and International credit mobility projects and other.

For more information about University of Montenegro, please visit our website www.ucg.ac.me or send e-mail to pr.centar@ac.me.

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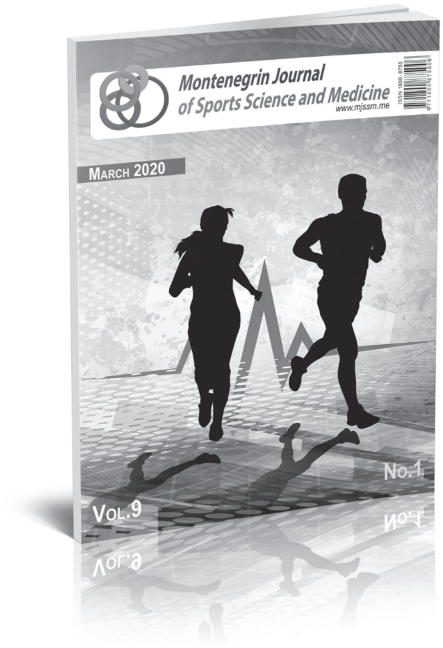

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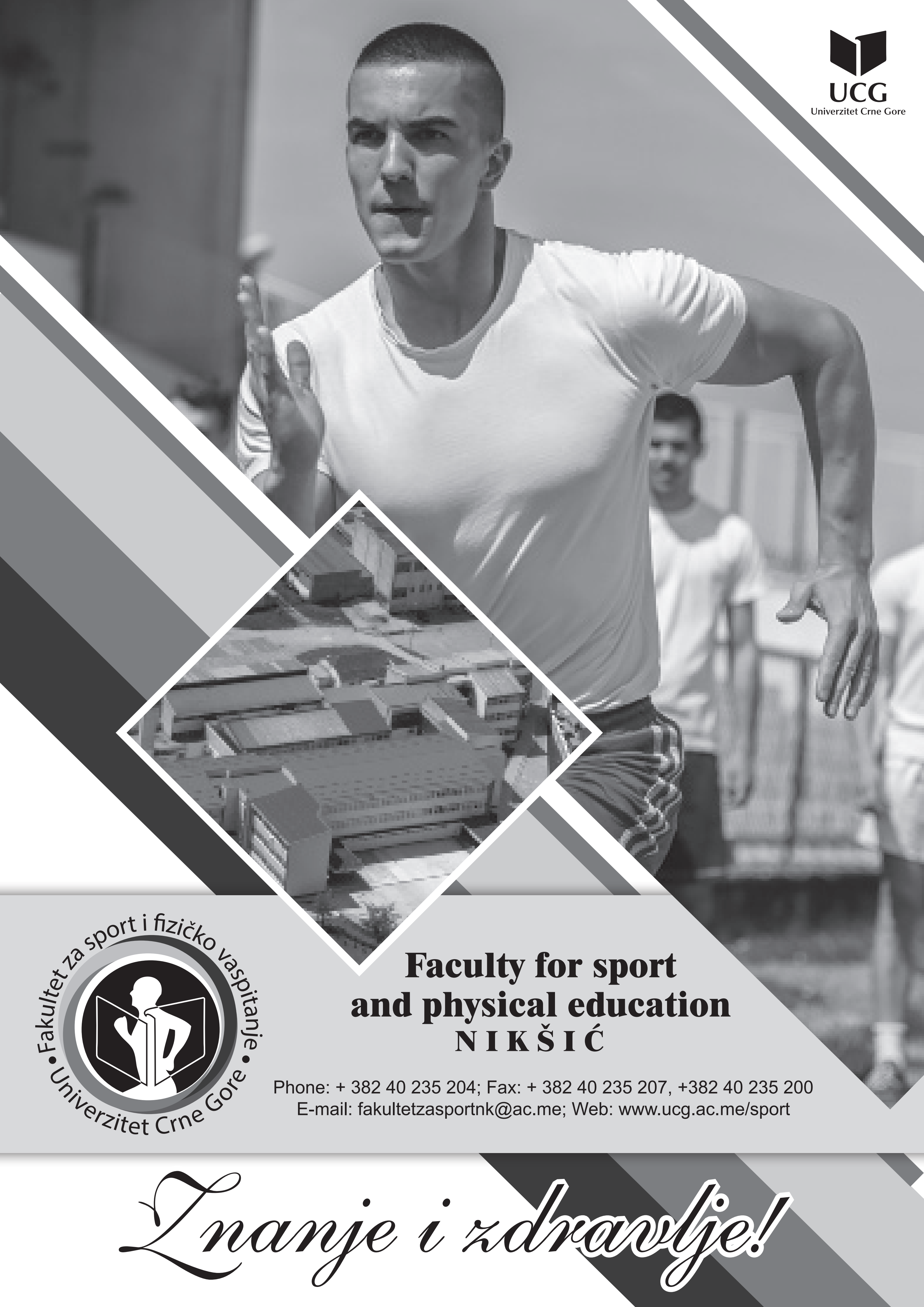
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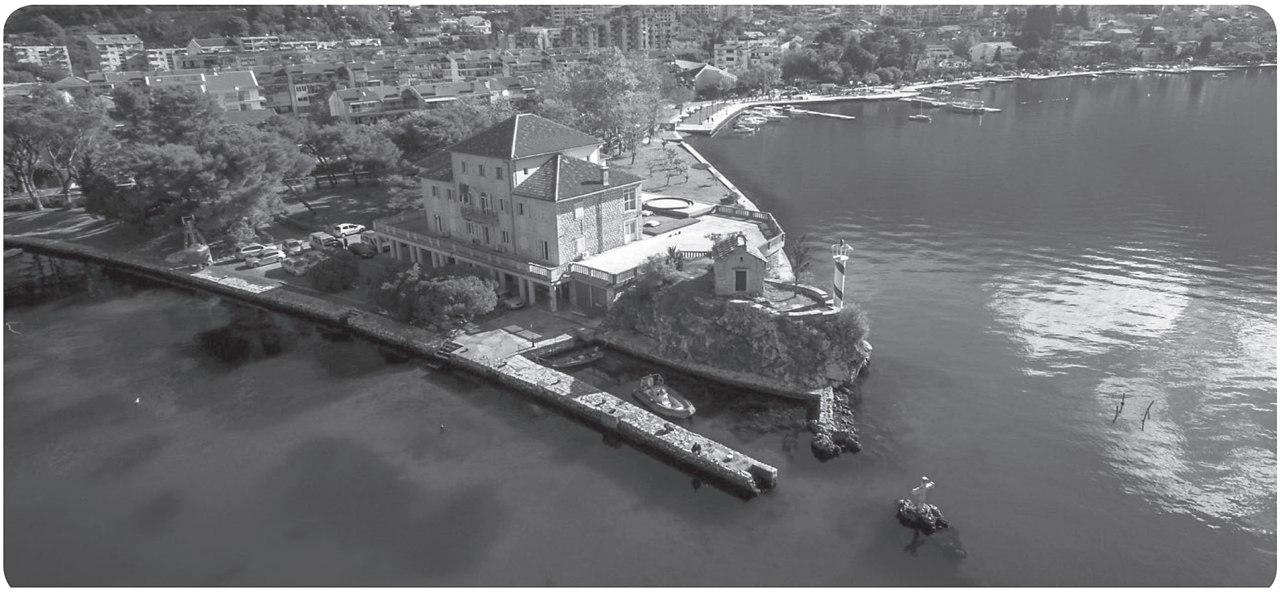
INSTITUT ZA BIOLOGIJU MORA



University of Montenegro – Institute for marine biology is located in Kotor, Montenegro. Since its establishment in 1961, the Institute performed comprehensive research of the marine and coastal area, which has its wide impact to the environmental protection, pollution-prevention and practical application. Core competencies of the Institute are focused on research in the fields of marine conservation, ichthyology and marine fisheries, marine chemistry, aquaculture, plankton research, neuro and eco-physiology. The main research area is investigating and protection of Adriatic sea with special interest of South Adriatic area. Institute for marine biology have a wide range of international cooperation with Marine research institutions and Universities all over Mediterranean area trough a numerous Eu funded scientific projects.

All over the year Institute is looking to hire a young students from the field of general biology, marine biology, marine chemistry, molecular biology or similar disciplines on voluntary basis to work with us. We need opportunity for international internship or MSc or PhD thesis that could be performed on Institute in our 5 different labs: Fisheries and ichthyology, Aquaculture, Marine chemistry, Plankton and sea water quality and Benthos and marine conservation.

Every year Institute organize several summer schools and workshop for interested students, MSc and PhD candidates. From 01-05 July 2019 we will organize Summer school "Blue Growth: emerging technologies, trends and opportunities" in frame of InnoBlueGrowth Project who is financed by Interreg Med programme. Through the specific theme courses, workshops and working labs offered – covering different areas of the blue economy – the Summer School aims at encouraging young people involvement in blue economy sectors by offering high-quality technical knowledge and fostering their entrepreneurial spirit. The Summer School will facilitate fruitful exchanges and a stronger understanding among a variety of actors coming from different Mediterranean countries with diverse profiles, including representatives from the academia, the public and private sectors, but also potential funders and investors. These activities will count on specific team building activities for participants as well to reinforce interpersonal skills and foster cohesion among blue academia and sectors.

If You are interested apply on the following link: https://www.ucg.ac.me/objava/blog/1221/objava/45392-ljetnja-skola-plavi-rast-nove-tehnologije-trendovi-i-mogucnosti

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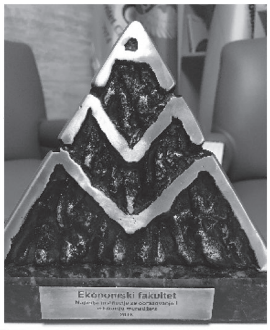
University of Montenegro
FACULTY of ECONOMICS PODGORICA

UCG
University of Montenegro

Faculty of Economics
University of Montenegro

The Faculty of Economics celebrated its 57th anniversary this year, and it is the oldest higher education institution in the country. Since its establishment, 8,630 students graduated at our Faculty.

Today, Faculty of Economics is a largely interdisciplinary institution, characterized by expressed dynamism in its work. Employees at the Faculty are dedicated to constant improvements and enhancements, all in accordance with the needs brought by the changes.



We provide our students with the best theoretical and practical knowledge, enabling them to develop critical spirit in approaching economic phenomena and solving concrete problems in daily work. From September 2017, at the Faculty, the new generation will start a 3 + 2 + 3 study, which will improve the quality of studying.

Development of Faculty of Economics in the coming period will follow the vision of development of the University of Montenegro, pursuing full achievement of its mission

Comprehensive literature, contemporary authors and works have always been imperative in creation of new academic directions at Faculty of Economics, which will form the basis of our future.

Faculty and its employees are dedicated to developing interest in strengthening the entrepreneurial initiative, creative and interdisciplinary approach among young people, using modern teaching and research methods. In this regard, the Faculty has modern textbooks and adequate IT technology, which supports the objectives set.

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